Final Project

Due December 13th at 12:00pm

Jordan Andrew, Pranshul Bhatnagar, Arushi Singh, and Jay Wu

Load Packages

```
library(tidyverse)
library(dplyr)
library(car)
library(ggplot2)
```

Load the data and provide a `glimpse()`:

```
library(dplyr)
music_data <- read.csv("data/mxmh_survey_results.csv")
glimpse(music_data)
```

```
Rows: 736
Columns: 33
$ Timestamp                    <chr> "8/27/2022 19:29:02", "8/27/2022 19:57:31~
$ Age                          <int> 18, 63, 18, 61, 18, 18, 18, 21, 19, 18, 1~
$ Primary.streaming.service    <chr> "Spotify", "Pandora", "Spotify", "YouTube~
$ Hours.per.day                <dbl> 3.0, 1.5, 4.0, 2.5, 4.0, 5.0, 3.0, 1.0, 6~
$ While.working                <chr> "Yes", "Yes", "No", "Yes", "Yes", "Yes", ~
$ Instrumentalist              <chr> "Yes", "No", "No", "No", "No", "Yes", "Ye~
$ Composer                     <chr> "Yes", "No", "No", "Yes", "No", "Yes", "N~
$ Fav.genre                    <chr> "Latin", "Rock", "Video game music", "Jaz~
$ Exploratory                  <chr> "Yes", "Yes", "No", "Yes", "Yes", "Yes", ~
$ Foreign.languages            <chr> "Yes", "No", "Yes", "Yes", "No", "Yes", "~
$ BPM                          <int> 156, 119, 132, 84, 107, 86, 66, 95, 94, 1~
$ Frequency..Classical.        <chr> "Rarely", "Sometimes", "Never", "Sometime~
$ Frequency..Country.          <chr> "Never", "Never", "Never", "Never", "Neve~
$ Frequency..EDM.              <chr> "Rarely", "Never", "Very frequently", "Ne~
$ Frequency..Folk.             <chr> "Never", "Rarely", "Never", "Rarely", "Ne~
```

```
$ Frequency..Gospel.           <chr> "Never", "Sometimes", "Never", "Sometimes~
$ Frequency..Hip.hop.          <chr> "Sometimes", "Rarely", "Rarely", "Never",~
$ Frequency..Jazz.             <chr> "Never", "Very frequently", "Rarely", "Ve~
$ Frequency..K.pop.            <chr> "Very frequently", "Rarely", "Very freque~
$ Frequency..Latin.            <chr> "Very frequently", "Sometimes", "Never", ~
$ Frequency..Lofi.             <chr> "Rarely", "Rarely", "Sometimes", "Sometim~
$ Frequency..Metal.            <chr> "Never", "Never", "Sometimes", "Never", "~
$ Frequency..Pop.              <chr> "Very frequently", "Sometimes", "Rarely",~
$ Frequency..R.B.              <chr> "Sometimes", "Sometimes", "Never", "Somet~
$ Frequency..Rap.              <chr> "Very frequently", "Rarely", "Rarely", "N~
$ Frequency..Rock.             <chr> "Never", "Very frequently", "Rarely", "Ne~
$ Frequency..Video.game.music. <chr> "Sometimes", "Rarely", "Very frequently",~
$ Anxiety                      <dbl> 3, 7, 7, 9, 7, 8, 4, 5, 2, 2, 7, 1, 9, 2,~
$ Depression                   <dbl> 0, 2, 7, 7, 2, 8, 8, 3, 0, 2, 7, 0, 3, 1,~
$ Insomnia                     <dbl> 1, 2, 10, 3, 5, 7, 6, 5, 0, 5, 4, 0, 2, 2~
$ OCD                          <dbl> 0, 1, 2, 3, 9, 7, 0, 3, 0, 1, 7, 1, 7, 0,~
$ Music.effects                <chr> "", "", "No effect", "Improve", "Improve"~
$ Permissions                  <chr> "I understand.", "I understand.", "I unde~
```

Data Cleaning

```
# Showing which columns have null values and how many
music_data %>%
  summarise(across(everything(), ~ sum(is.na(.)))) %>%
  pivot_longer(everything(), names_to = "column", values_to = "na_count") %>%
  filter(na_count > 0) %>%
  arrange(desc(na_count))
```

```
# A tibble: 2 x 2
  column na_count
  <chr>     <int>
1 BPM         107
2 Age           1
```

```
# Summary to spot outliers
summary(music_data$Age)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.    NA's
  10.00   18.00   21.00   25.21   28.00   89.00       1
```

```r
# Count unrealistic ages
sum(music_data$Age < 10 | music_data$Age > 90, na.rm = TRUE)
```

```
[1] 0
```

```r
# View rows with unrealistic ages
music_data %>%
  filter(Age < 10 | Age > 90)
```

```
 [1] Timestamp                    Age
 [3] Primary.streaming.service    Hours.per.day
 [5] While.working                Instrumentalist
 [7] Composer                     Fav.genre
 [9] Exploratory                  Foreign.languages
[11] BPM                          Frequency..Classical.
[13] Frequency..Country.          Frequency..EDM.
[15] Frequency..Folk.             Frequency..Gospel.
[17] Frequency..Hip.hop.          Frequency..Jazz.
[19] Frequency..K.pop.            Frequency..Latin.
[21] Frequency..Lofi.             Frequency..Metal.
[23] Frequency..Pop.              Frequency..R.B.
[25] Frequency..Rap.              Frequency..Rock.
[27] Frequency..Video.game.music. Anxiety
[29] Depression                   Insomnia
[31] OCD                          Music.effects
[33] Permissions
<0 rows> (or 0-length row.names)
```

```r
# Checking that music hours per day doesn't contain unrealistic # of hours
summary(music_data$Hours.per.day)
```

```
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  0.000   2.000   3.000   3.573   5.000  24.000
```

```r
music_data %>%
  filter(Hours.per.day < 0 | Hours.per.day > 16)
```

```
           Timestamp Age Primary.streaming.service Hours.per.day While.working
1  8/27/2022 23:40:55  17                   Spotify            24           Yes
```

```
2   8/29/2022 2:46:27  27                   Spotify            20           Yes
3  8/29/2022 12:32:30  16                   Spotify            24           Yes
4  9/28/2022 17:25:48  89                   Spotify            24           Yes
5 10/23/2022 20:50:27  18               Apple Music            18           Yes
  Instrumentalist Composer Fav.genre Exploratory Foreign.languages BPM
1              No       No       Rap         Yes                No  99
2              No       No      Rock         Yes               Yes  NA
3              No       No      Rock         Yes               Yes 120
4             Yes      Yes       Rap          No                No 143
5              No       No       EDM         Yes                No  90
  Frequency..Classical. Frequency..Country. Frequency..EDM. Frequency..Folk.
1                Rarely               Never           Never            Never
2             Sometimes              Rarely       Sometimes           Rarely
3                 Never               Never          Rarely            Never
4                 Never               Never          Rarely           Rarely
5             Sometimes              Rarely Very frequently            Never
  Frequency..Gospel. Frequency..Hip.hop. Frequency..Jazz. Frequency..K.pop.
1              Never           Sometimes        Sometimes         Sometimes
2              Never              Rarely            Never         Sometimes
3              Never               Never            Never            Rarely
4              Never     Very frequently        Sometimes             Never
5             Rarely           Sometimes        Sometimes             Never
  Frequency..Latin. Frequency..Lofi. Frequency..Metal. Frequency..Pop.
1            Rarely            Never         Sometimes          Rarely
2            Rarely        Sometimes   Very frequently          Rarely
3            Rarely           Rarely            Rarely          Rarely
4             Never           Rarely            Rarely          Rarely
5             Never        Sometimes         Sometimes          Rarely
  Frequency..R.B. Frequency..Rap. Frequency..Rock. Frequency..Video.game.music.
1       Sometimes Very frequently  Very frequently                        Never
2           Never          Rarely  Very frequently                    Sometimes
3          Rarely          Rarely        Sometimes                       Rarely
4       Sometimes Very frequently           Rarely                        Never
5       Sometimes       Sometimes        Sometimes                    Sometimes
  Anxiety Depression Insomnia OCD Music.effects   Permissions
1       7          5        0   3       Improve I understand.
2       8         10       10   8     No effect I understand.
3       5          1        9   1       Improve I understand.
4       0          0        0   0     No effect I understand.
5       9          8        5  10       Improve I understand.
```

```r
# we will remove the 89 year old who listens to rap exclusively while working and has no perc
```

```r
cols <- c("Anxiety", "Depression", "Insomnia", "OCD")

# Checking if any values for self reported mental illness are out of the 0-10 range
music_data %>%
  summarise(across(all_of(cols),
                   ~ sum(. < 0 | . > 10, na.rm = TRUE)))
```

```
  Anxiety Depression Insomnia OCD
1       0          0        0   0
```

```r
yn_cols <- c(
  "While.working",
  "Instrumentalist",
  "Composer",
  "Exploratory",
  "Foreign.languages"
)

# Checking to see if any of the "yes" "no" columns have values outside of these options
sapply(music_data[yn_cols], function(x) all(x %in% c("Yes", "No")))
```

```
    While.working   Instrumentalist          Composer       Exploratory
            FALSE             FALSE             FALSE              TRUE
Foreign.languages
            FALSE
```

```r
lapply(music_data[yn_cols], function(x) setdiff(unique(x), c("Yes", "No")))
```

```
$While.working
[1] ""

$Instrumentalist
[1] ""

$Composer
[1] ""

$Exploratory
```

```
character(0)

$Foreign.languages
[1] ""
```

```r
sapply(music_data[yn_cols], function(x) sum(x == "", na.rm = TRUE))
```

```
    While.working   Instrumentalist          Composer       Exploratory
                3                 4                 1                 0
Foreign.languages
                4
```

```r
# We will drop the rows that having missing values here, as there is no clear way how to impu
```

```r
freq_cols <- c(
  "Frequency..Classical.",
  "Frequency..Country.",
  "Frequency..EDM.",
  "Frequency..Folk.",
  "Frequency..Gospel.",
  "Frequency..Hip.hop.",
  "Frequency..Jazz.",
  "Frequency..K.pop.",
  "Frequency..Latin.",
  "Frequency..Lofi.",
  "Frequency..Metal.",
  "Frequency..Pop.",
  "Frequency..R.B.",
  "Frequency..Rap.",
  "Frequency..Rock.",
  "Frequency..Video.game.music."
)

allowed_freq <- c("Never", "Rarely", "Sometimes", "Very frequently")


# Checking to see if any of the frequency columns have values outside of "Never, Rarely, Some
sapply(music_data[freq_cols], function(x) sum(!(x %in% allowed_freq)))
```

```
       Frequency..Classical.          Frequency..Country.
                           0                            0
```

```
      Frequency..EDM.            Frequency..Folk.
                    0                           0
   Frequency..Gospel.         Frequency..Hip.hop.
                    0                           0
     Frequency..Jazz.           Frequency..K.pop.
                    0                           0
    Frequency..Latin.            Frequency..Lofi.
                    0                           0
    Frequency..Metal.             Frequency..Pop.
                    0                           0
      Frequency..R.B.             Frequency..Rap.
                    0                           0
     Frequency..Rock. Frequency..Video.game.music.
                    0                           0
```

```
table(music_data$Music.effects)
```

```
        Improve No effect    Worsen
        8       542       169        17
```

General Data Cleaning

```
# Cleaning our data

# dropping BPM column
music_data <- music_data %>% select(-any_of("BPM"))

# dropping 1 row that has a null value for Age
music_data <- music_data %>%
  filter(!is.na(Age))

# Dropping the 89 year old year old who listens to only rap while working 24 hours a day(unre
music_data <- music_data %>%
  filter(!(Age == 89 & Hours.per.day == 24))

# Dropping the rows that have non "yes" "no" answers for questions that only permit those val
allowed_yn <- c("Yes", "No")
```

```r
music_data <- music_data %>%
  filter(
    if_all(all_of(yn_cols), ~ .x %in% allowed_yn)
  )

# removing 7 rows where Music.effects has an empty string as a response
music_data <- music_data %>%
  filter(Music.effects != "")

# Keep only columns that exist in the data
freq_cols_present <- freq_cols[freq_cols %in% colnames(music_data)]

# Only rename if any of these columns exist
if (length(freq_cols_present) > 0) {
  music_data <- music_data %>%
    rename_with(.cols = all_of(freq_cols_present),
                .fn = ~ gsub("Frequency\\.\\.", "freq_", .x) %>%
                       gsub("\\.", "", .))  # Remove all remaining periods
}

music_data_clean <- music_data

glimpse(music_data_clean)
```

```
Rows: 718
Columns: 32
$ Timestamp                    <chr> "8/27/2022 21:28:18", "8/27/2022 21:40:40", ~
$ Age                          <int> 18, 61, 18, 18, 18, 21, 19, 18, 18, 19, 19, ~
$ Primary.streaming.service <chr> "Spotify", "YouTube Music", "Spotify", "Spot~
$ Hours.per.day                <dbl> 4.0, 2.5, 4.0, 5.0, 3.0, 1.0, 6.0, 1.0, 3.0,~
$ While.working                <chr> "No", "Yes", "Yes", "Yes", "Yes", "Yes", "Ye~
$ Instrumentalist              <chr> "No", "No", "No", "Yes", "Yes", "No", "No", ~
$ Composer                     <chr> "No", "Yes", "No", "Yes", "No", "No", "No", ~
$ Fav.genre                    <chr> "Video game music", "Jazz", "R&B", "Jazz", "~
$ Exploratory                  <chr> "No", "Yes", "Yes", "Yes", "Yes", "Yes", "No~
$ Foreign.languages            <chr> "Yes", "Yes", "No", "Yes", "Yes", "Yes", "No~
$ freq_Classical               <chr> "Never", "Sometimes", "Never", "Rarely", "So~
$ freq_Country                 <chr> "Never", "Never", "Never", "Sometimes", "Nev~
$ freq_EDM                     <chr> "Very frequently", "Never", "Rarely", "Never~
$ freq_Folk                    <chr> "Never", "Rarely", "Never", "Never", "Someti~
$ freq_Gospel                  <chr> "Never", "Sometimes", "Rarely", "Never", "Ra~
$ freq_Hiphop                  <chr> "Rarely", "Never", "Very frequently", "Somet~
```

```
$ freq_Jazz                <chr> "Rarely", "Very frequently", "Never", "Very ~
$ freq_Kpop                <chr> "Very frequently", "Sometimes", "Very freque~
$ freq_Latin               <chr> "Never", "Very frequently", "Sometimes", "Ra~
$ freq_Lofi                <chr> "Sometimes", "Sometimes", "Sometimes", "Very~
$ freq_Metal               <chr> "Sometimes", "Never", "Never", "Rarely", "Ra~
$ freq_Pop                 <chr> "Rarely", "Sometimes", "Sometimes", "Very fr~
$ freq_RB                  <chr> "Never", "Sometimes", "Very frequently", "Ve~
$ freq_Rap                 <chr> "Rarely", "Never", "Very frequently", "Very ~
$ freq_Rock                <chr> "Rarely", "Never", "Never", "Very frequently~
$ freq_Videogamemusic      <chr> "Very frequently", "Never", "Rarely", "Never~
$ Anxiety                  <dbl> 7, 9, 7, 8, 4, 5, 2, 2, 7, 1, 2, 6, 7, 8, 5,~
$ Depression               <dbl> 7, 7, 2, 8, 8, 3, 0, 2, 7, 0, 1, 4, 5, 8, 7,~
$ Insomnia                 <dbl> 10, 3, 5, 7, 6, 5, 0, 5, 4, 0, 2, 7, 4, 4, 1~
$ OCD                      <dbl> 2, 3, 9, 7, 0, 3, 0, 1, 7, 1, 0, 0, 1, 3, 0,~
$ Music.effects            <chr> "No effect", "Improve", "Improve", "Improve"~
$ Permissions              <chr> "I understand.", "I understand.", "I underst~
```

Research Question 1

```
# Question 1: How does the number of hours spent listening to music per day vary by depressi

# this will be the dataset used for our first research question
music_data_1 <- music_data_clean

unique(music_data_1$Primary.streaming.service)
```

```
[1] "Spotify"                           "YouTube Music"
[3] "I do not use a streaming service." "Apple Music"
[5] "Other streaming service"           "Pandora"
[7] ""
```

```
music_data_1 %>%
  count(Primary.streaming.service)
```

```
          Primary.streaming.service   n
1                                     1
2                       Apple Music  50
3 I do not use a streaming service.  69
4           Other streaming service  49
```

```
5                           Pandora  10
6                           Spotify 449
7                     YouTube Music  90
```

```r
# removing the row where Primary.streaming.service is empty
music_data_1 <- music_data_1 %>%
  filter(Primary.streaming.service != "")

music_data_1 %>%
  filter(Primary.streaming.service == "I do not use a streaming service.")
```

```
             Timestamp Age         Primary.streaming.service Hours.per.day
1   8/27/2022 22:44:03  18 I do not use a streaming service.          1.00
2    8/28/2022 1:39:02  19 I do not use a streaming service.          5.00
3   8/28/2022 12:12:35  17 I do not use a streaming service.          1.00
4   8/28/2022 13:18:19  23 I do not use a streaming service.          1.00
5   8/28/2022 13:57:47  36 I do not use a streaming service.          2.00
6   8/28/2022 14:30:34  33 I do not use a streaming service.          1.00
7   8/28/2022 14:47:08  43 I do not use a streaming service.          6.00
8   8/28/2022 15:22:00  25 I do not use a streaming service.          0.50
9   8/28/2022 18:35:17  21 I do not use a streaming service.          3.00
10  8/28/2022 18:56:54  28 I do not use a streaming service.          4.00
11  8/28/2022 19:29:08  16 I do not use a streaming service.          2.00
12  8/28/2022 19:50:30  21 I do not use a streaming service.          2.00
13  8/28/2022 21:23:07  23 I do not use a streaming service.          6.00
14  8/28/2022 21:55:22  36 I do not use a streaming service.          8.00
15  8/28/2022 22:14:30  64 I do not use a streaming service.          4.00
16  8/28/2022 22:51:01  53 I do not use a streaming service.          1.50
17  8/28/2022 23:54:34  15 I do not use a streaming service.          5.00
18  8/28/2022 23:56:06  17 I do not use a streaming service.          0.50
19   8/29/2022 0:33:06  18 I do not use a streaming service.          2.00
20   8/29/2022 0:47:12  25 I do not use a streaming service.          1.00
21   8/29/2022 1:22:44  24 I do not use a streaming service.          8.00
22   8/29/2022 1:54:06  20 I do not use a streaming service.          1.50
23   8/29/2022 3:09:21  15 I do not use a streaming service.          5.00
24   8/29/2022 3:49:29  56 I do not use a streaming service.          1.00
25   8/29/2022 4:42:51  32 I do not use a streaming service.          1.00
26   8/29/2022 8:39:19  21 I do not use a streaming service.         10.00
27   8/29/2022 8:47:09  22 I do not use a streaming service.          2.00
28  8/29/2022 11:04:24  20 I do not use a streaming service.          0.50
29  8/29/2022 11:41:34  29 I do not use a streaming service.          2.00
30  8/29/2022 16:39:08  32 I do not use a streaming service.          3.00
```

31	8/29/2022 17:32:02	20	I do not use a streaming service.	2.00
32	8/29/2022 17:36:01	23	I do not use a streaming service.	2.00
33	8/29/2022 18:56:27	54	I do not use a streaming service.	6.00
34	8/30/2022 17:33:13	31	I do not use a streaming service.	4.00
35	9/1/2022 9:25:07	20	I do not use a streaming service.	4.00
36	9/1/2022 14:59:20	33	I do not use a streaming service.	4.00
37	9/1/2022 15:01:10	22	I do not use a streaming service.	2.00
38	9/1/2022 15:24:02	34	I do not use a streaming service.	2.00
39	9/1/2022 16:00:05	23	I do not use a streaming service.	10.00
40	9/1/2022 17:17:48	70	I do not use a streaming service.	2.00
41	9/1/2022 19:44:33	71	I do not use a streaming service.	0.25
42	9/1/2022 20:36:10	18	I do not use a streaming service.	1.00
43	9/1/2022 21:07:33	74	I do not use a streaming service.	1.00
44	9/1/2022 23:11:19	19	I do not use a streaming service.	7.00
45	9/2/2022 4:56:56	51	I do not use a streaming service.	5.00
46	9/2/2022 7:29:17	19	I do not use a streaming service.	1.00
47	9/2/2022 9:25:05	18	I do not use a streaming service.	4.00
48	9/2/2022 14:09:04	80	I do not use a streaming service.	3.00
49	9/2/2022 14:31:06	22	I do not use a streaming service.	4.00
50	9/3/2022 3:54:07	50	I do not use a streaming service.	1.00
51	9/3/2022 18:37:51	22	I do not use a streaming service.	2.00
52	9/3/2022 21:08:12	25	I do not use a streaming service.	4.00
53	9/5/2022 12:10:54	26	I do not use a streaming service.	1.00
54	9/6/2022 14:43:07	23	I do not use a streaming service.	10.00
55	9/9/2022 11:15:23	46	I do not use a streaming service.	1.00
56	9/9/2022 23:34:31	19	I do not use a streaming service.	0.00
57	9/10/2022 9:06:07	56	I do not use a streaming service.	2.00
58	9/12/2022 14:29:16	48	I do not use a streaming service.	5.00
59	9/12/2022 15:07:48	21	I do not use a streaming service.	1.00
60	9/12/2022 19:13:06	49	I do not use a streaming service.	8.00
61	9/12/2022 22:26:06	30	I do not use a streaming service.	0.00
62	9/13/2022 11:16:37	43	I do not use a streaming service.	0.00
63	9/13/2022 13:18:32	24	I do not use a streaming service.	12.00
64	9/13/2022 13:58:18	21	I do not use a streaming service.	0.50
65	9/13/2022 16:16:16	28	I do not use a streaming service.	1.00
66	9/23/2022 23:06:29	18	I do not use a streaming service.	1.00
67	9/23/2022 23:43:10	18	I do not use a streaming service.	1.00
68	10/4/2022 8:53:39	23	I do not use a streaming service.	3.00
69	10/30/2022 13:13:32	21	I do not use a streaming service.	0.50

	While.working	Instrumentalist	Composer	Fav.genre	Exploratory
1	Yes	No	No	R&B	Yes
2	Yes	No	No	R&B	Yes
3	Yes	Yes	Yes	Folk	No

4	Yes	Yes	No	Video game music	No
5	Yes	Yes	Yes	Rock	Yes
6	No	No	No	Metal	No
7	Yes	Yes	Yes	Video game music	Yes
8	No	Yes	No	Classical	No
9	Yes	No	No	Rock	No
10	Yes	No	No	Rock	Yes
11	No	Yes	Yes	Video game music	Yes
12	Yes	No	No	Rock	No
13	Yes	No	Yes	Classical	No
14	Yes	Yes	Yes	Video game music	No
15	Yes	No	No	Rock	No
16	No	No	No	Pop	Yes
17	Yes	No	No	Rock	Yes
18	Yes	Yes	No	Video game music	No
19	Yes	Yes	No	Metal	Yes
20	No	No	No	Pop	No
21	Yes	No	Yes	Video game music	Yes
22	No	Yes	Yes	Video game music	No
23	Yes	Yes	No	Metal	Yes
24	Yes	No	No	Pop	No
25	No	No	No	Rock	No
26	Yes	No	No	Pop	Yes
27	Yes	No	No	Video game music	No
28	No	No	No	Metal	Yes
29	Yes	No	No	Metal	Yes
30	Yes	Yes	No	Rock	No
31	Yes	No	No	Rock	Yes
32	Yes	No	No	Pop	Yes
33	Yes	No	No	Classical	No
34	Yes	Yes	No	Classical	Yes
35	Yes	No	No	Rock	Yes
36	Yes	Yes	No	Rock	Yes
37	No	Yes	Yes	Metal	No
38	No	No	No	Video game music	No
39	Yes	No	No	Video game music	No
40	Yes	No	No	Country	No
41	Yes	No	No	Gospel	No
42	Yes	No	Yes	Video game music	No
43	No	No	No	Pop	No
44	Yes	No	No	Rock	No
45	Yes	No	No	Rock	No
46	Yes	Yes	No	Video game music	No

```
47          Yes          Yes    No              EDM          No
48          Yes          Yes    No        Classical          No
49           No           No   Yes              Pop         Yes
50           No           No    No             Rock          No
51          Yes           No    No Video game music          No
52          Yes           No    No              Pop         Yes
53           No           No    No              Pop          No
54          Yes          Yes   Yes            Metal          No
55           No           No    No             Rock          No
56           No           No    No        Classical         Yes
57          Yes           No    No             Rock          No
58          Yes           No   Yes             Rock         Yes
59          Yes          Yes    No        Classical          No
60          Yes          Yes   Yes        Classical          No
61           No           No    No            Metal          No
62           No           No    No             Rock          No
63          Yes          Yes   Yes             Rock         Yes
64           No           No    No              Pop          No
65          Yes           No    No             Rock          No
66          Yes           No    No        Classical          No
67           No          Yes    No        Classical          No
68          Yes           No    No             Rock          No
69           No           No    No              Pop         Yes
   Foreign.languages  freq_Classical    freq_Country        freq_EDM
1                Yes          Rarely          Rarely          Rarely
2                Yes          Rarely          Rarely       Sometimes
3                 No           Never       Sometimes          Rarely
4                 No       Sometimes       Sometimes       Sometimes
5                 No       Sometimes          Rarely Very frequently
6                Yes          Rarely       Sometimes       Sometimes
7                Yes       Sometimes          Rarely Very frequently
8                 No Very frequently           Never           Never
9                 No           Never           Never           Never
10               Yes          Rarely       Sometimes          Rarely
11                No       Sometimes          Rarely           Never
12               Yes          Rarely       Sometimes          Rarely
13               Yes Very frequently           Never           Never
14               Yes       Sometimes          Rarely       Sometimes
15                No       Sometimes          Rarely       Sometimes
16               Yes          Rarely           Never Very frequently
17               Yes          Rarely           Never Very frequently
18               Yes          Rarely       Sometimes       Sometimes
19               Yes       Sometimes           Never          Rarely
```

20	Yes	Rarely	Never	Never
21	No	Sometimes	Rarely	Very frequently
22	Yes	Very frequently	Never	Never
23	No	Never	Never	Rarely
24	No	Never	Rarely	Never
25	No	Rarely	Sometimes	Never
26	No	Never	Rarely	Never
27	Yes	Never	Never	Rarely
28	Yes	Sometimes	Rarely	Sometimes
29	No	Never	Never	Never
30	Yes	Very frequently	Sometimes	Sometimes
31	No	Rarely	Rarely	Never
32	Yes	Rarely	Very frequently	Never
33	No	Very frequently	Sometimes	Never
34	Yes	Very frequently	Rarely	Never
35	Yes	Rarely	Never	Sometimes
36	Yes	Rarely	Sometimes	Rarely
37	No	Never	Never	Rarely
38	Yes	Rarely	Rarely	Rarely
39	Yes	Rarely	Never	Sometimes
40	No	Sometimes	Very frequently	Never
41	Yes	Sometimes	Rarely	Never
42	No	Rarely	Rarely	Sometimes
43	No	Rarely	Very frequently	Never
44	No	Sometimes	Rarely	Never
45	No	Sometimes	Rarely	Never
46	No	Very frequently	Never	Never
47	Yes	Very frequently	Never	Very frequently
48	No	Very frequently	Rarely	Never
49	No	Rarely	Sometimes	Very frequently
50	No	Sometimes	Sometimes	Never
51	No	Sometimes	Rarely	Rarely
52	Yes	Rarely	Never	Sometimes
53	No	Never	Rarely	Sometimes
54	Yes	Never	Never	Never
55	No	Never	Sometimes	Never
56	Yes	Very frequently	Rarely	Sometimes
57	Yes	Rarely	Never	Never
58	Yes	Very frequently	Sometimes	Sometimes
59	No	Very frequently	Never	Never
60	No	Very frequently	Never	Never
61	No	Rarely	Never	Rarely
62	No	Rarely	Never	Never

```
63                No       Sometimes          Rarely           Never
64                No           Never           Never           Never
65                No          Rarely          Rarely           Never
66                No       Sometimes       Sometimes          Rarely
67                No Very frequently           Never       Sometimes
68                No           Never           Never           Never
69                No           Never          Rarely       Sometimes
         freq_Folk     freq_Gospel     freq_Hiphop       freq_Jazz
1           Rarely       Sometimes          Rarely          Rarely
2  Very frequently          Rarely Very frequently       Sometimes
3  Very frequently           Never          Rarely           Never
4           Rarely           Never          Rarely       Sometimes
5        Sometimes          Rarely          Rarely Very frequently
6            Never           Never       Sometimes           Never
7           Rarely          Rarely          Rarely          Rarely
8            Never          Rarely           Never       Sometimes
9        Sometimes       Sometimes       Sometimes       Sometimes
10          Rarely           Never       Sometimes       Sometimes
11           Never          Rarely       Sometimes       Sometimes
12       Sometimes           Never Very frequently          Rarely
13          Rarely           Never           Never           Never
14          Rarely           Never          Rarely          Rarely
15          Rarely           Never          Rarely       Sometimes
16       Sometimes           Never       Sometimes           Never
17          Rarely           Never           Never           Never
18           Never           Never           Never          Rarely
19           Never           Never           Never           Never
20          Rarely           Never          Rarely          Rarely
21          Rarely          Rarely Very frequently Very frequently
22          Rarely          Rarely           Never           Never
23           Never           Never           Never           Never
24          Rarely           Never           Never           Never
25       Sometimes           Never           Never           Never
26           Never           Never          Rarely           Never
27           Never           Never       Sometimes          Rarely
28           Never           Never       Sometimes       Sometimes
29           Never           Never           Never           Never
30       Sometimes           Never           Never Very frequently
31       Sometimes           Never           Never          Rarely
32          Rarely Very frequently Very frequently       Sometimes
33           Never           Never       Sometimes          Rarely
34          Rarely          Rarely           Never       Sometimes
35           Never           Never          Rarely           Never
```

36	Rarely	Rarely	Rarely	Rarely
37	Never	Never	Never	Never
38	Never	Never	Sometimes	Rarely
39	Sometimes	Never	Rarely	Never
40	Never	Never	Never	Never
41	Rarely	Sometimes	Never	Never
42	Sometimes	Never	Rarely	Sometimes
43	Sometimes	Sometimes	Never	Rarely
44	Rarely	Never	Never	Sometimes
45	Sometimes	Never	Never	Rarely
46	Rarely	Rarely	Sometimes	Sometimes
47	Never	Never	Never	Rarely
48	Rarely	Never	Never	Rarely
49	Sometimes	Never	Rarely	Never
50	Never	Never	Very frequently	Sometimes
51	Never	Very frequently	Rarely	Sometimes
52	Never	Never	Very frequently	Never
53	Sometimes	Never	Rarely	Rarely
54	Never	Never	Never	Never
55	Never	Never	Never	Never
56	Rarely	Rarely	Sometimes	Rarely
57	Never	Never	Never	Never
58	Very frequently	Never	Sometimes	Rarely
59	Never	Never	Never	Never
60	Never	Never	Never	Rarely
61	Never	Never	Never	Never
62	Never	Never	Rarely	Never
63	Sometimes	Rarely	Rarely	Rarely
64	Never	Never	Never	Never
65	Rarely	Never	Never	Never
66	Rarely	Rarely	Sometimes	Sometimes
67	Never	Never	Sometimes	Rarely
68	Rarely	Never	Never	Sometimes
69	Never	Never	Sometimes	Rarely

	freq_Kpop	freq_Latin	freq_Lofi	freq_Metal
1	Never	Rarely	Rarely	Never
2	Never	Never	Rarely	Rarely
3	Never	Rarely	Rarely	Rarely
4	Rarely	Rarely	Very frequently	Never
5	Never	Never	Never	Very frequently
6	Never	Rarely	Never	Very frequently
7	Sometimes	Rarely	Very frequently	Rarely
8	Never	Sometimes	Never	Rarely

9	Never	Rarely	Never	Rarely
10	Rarely	Never	Never	Sometimes
11	Rarely	Never	Sometimes	Never
12	Never	Very frequently	Very frequently	Very frequently
13	Never	Rarely	Never	Never
14	Never	Never	Never	Very frequently
15	Rarely	Rarely	Never	Rarely
16	Rarely	Sometimes	Never	Never
17	Never	Never	Sometimes	Rarely
18	Never	Never	Never	Rarely
19	Never	Never	Never	Very frequently
20	Never	Never	Rarely	Rarely
21	Rarely	Rarely	Very frequently	Very frequently
22	Never	Sometimes	Never	Never
23	Never	Never	Never	Very frequently
24	Never	Never	Never	Never
25	Never	Rarely	Never	Sometimes
26	Never	Never	Never	Never
27	Never	Rarely	Never	Never
28	Never	Never	Sometimes	Sometimes
29	Never	Never	Never	Very frequently
30	Rarely	Rarely	Never	Very frequently
31	Never	Never	Never	Rarely
32	Rarely	Never	Rarely	Never
33	Rarely	Rarely	Never	Rarely
34	Never	Never	Rarely	Rarely
35	Very frequently	Rarely	Rarely	Sometimes
36	Never	Never	Very frequently	Rarely
37	Never	Never	Never	Very frequently
38	Rarely	Never	Rarely	Rarely
39	Sometimes	Never	Sometimes	Rarely
40	Never	Never	Never	Never
41	Never	Sometimes	Never	Never
42	Never	Never	Never	Sometimes
43	Never	Sometimes	Never	Never
44	Never	Never	Never	Rarely
45	Never	Never	Never	Sometimes
46	Rarely	Rarely	Very frequently	Never
47	Never	Never	Rarely	Never
48	Rarely	Never	Rarely	Rarely
49	Never	Never	Never	Rarely
50	Rarely	Rarely	Never	Rarely
51	Rarely	Rarely	Rarely	Never

```
52          Rarely          Rarely          Rarely       Sometimes
53       Sometimes       Sometimes Very frequently           Never
54           Never           Never           Never Very frequently
55           Never           Never           Never           Never
56          Rarely          Rarely          Rarely           Never
57           Never           Never           Never Very frequently
58           Never          Rarely           Never          Rarely
59           Never           Never           Never           Never
60           Never           Never           Never          Rarely
61           Never           Never           Never       Sometimes
62           Never           Never           Never           Never
63           Never          Rarely           Never Very frequently
64       Sometimes           Never           Never           Never
65           Never           Never           Never           Never
66          Rarely          Rarely Very frequently           Never
67           Never           Never           Never           Never
68           Never           Never          Rarely           Never
69       Sometimes           Never Very frequently           Never
          freq_Pop         freq_RB        freq_Rap       freq_Rock
1        Sometimes       Sometimes          Rarely       Sometimes
2           Rarely Very frequently Very frequently       Sometimes
3           Rarely          Rarely       Sometimes       Sometimes
4        Sometimes Very frequently          Rarely       Sometimes
5        Sometimes          Rarely          Rarely Very frequently
6        Sometimes          Rarely       Sometimes       Sometimes
7        Sometimes          Rarely           Never       Sometimes
8           Rarely          Rarely           Never Very frequently
9           Rarely          Rarely       Sometimes Very frequently
10       Sometimes          Rarely          Rarely Very frequently
11       Sometimes       Sometimes          Rarely          Rarely
12 Very frequently Very frequently Very frequently Very frequently
13          Rarely           Never           Never           Never
14          Rarely           Never          Rarely       Sometimes
15       Sometimes          Rarely          Rarely       Sometimes
16 Very frequently          Rarely       Sometimes       Sometimes
17 Very frequently           Never           Never Very frequently
18       Sometimes          Rarely          Rarely       Sometimes
19          Rarely           Never          Rarely Very frequently
20 Very frequently          Rarely          Rarely Very frequently
21          Rarely          Rarely       Sometimes Very frequently
22           Never           Never           Never           Never
23          Rarely           Never          Rarely       Sometimes
24       Sometimes       Sometimes           Never       Sometimes
```

25	Very frequently	Rarely	Never	Very frequently
26	Sometimes	Rarely	Never	Sometimes
27	Very frequently	Never	Rarely	Sometimes
28	Rarely	Rarely	Rarely	Sometimes
29	Sometimes	Never	Never	Very frequently
30	Rarely	Never	Rarely	Very frequently
31	Sometimes	Rarely	Never	Very frequently
32	Very frequently	Very frequently	Very frequently	Very frequently
33	Sometimes	Sometimes	Rarely	Rarely
34	Never	Never	Never	Sometimes
35	Very frequently	Rarely	Rarely	Very frequently
36	Sometimes	Rarely	Rarely	Very frequently
37	Rarely	Rarely	Never	Very frequently
38	Never	Never	Sometimes	Rarely
39	Sometimes	Never	Rarely	Sometimes
40	Never	Never	Never	Never
41	Rarely	Rarely	Never	Rarely
42	Never	Rarely	Sometimes	Sometimes
43	Very frequently	Sometimes	Never	Sometimes
44	Never	Never	Never	Very frequently
45	Rarely	Rarely	Never	Very frequently
46	Sometimes	Sometimes	Rarely	Rarely
47	Sometimes	Never	Rarely	Rarely
48	Sometimes	Never	Never	Never
49	Very frequently	Very frequently	Rarely	Sometimes
50	Very frequently	Sometimes	Sometimes	Very frequently
51	Rarely	Never	Never	Never
52	Very frequently	Rarely	Very frequently	Very frequently
53	Very frequently	Rarely	Rarely	Sometimes
54	Never	Never	Never	Sometimes
55	Sometimes	Never	Never	Sometimes
56	Rarely	Rarely	Rarely	Rarely
57	Never	Never	Rarely	Very frequently
58	Rarely	Rarely	Rarely	Very frequently
59	Never	Never	Never	Never
60	Rarely	Never	Never	Rarely
61	Never	Never	Rarely	Rarely
62	Sometimes	Rarely	Rarely	Sometimes
63	Very frequently	Rarely	Rarely	Very frequently
64	Sometimes	Never	Never	Never
65	Sometimes	Never	Never	Sometimes
66	Rarely	Rarely	Rarely	Rarely
67	Sometimes	Never	Rarely	Never

```
68       Sometimes         Rarely          Never Very frequently
69 Very frequently      Sometimes      Sometimes Very frequently
   freq_Videogamemusic Anxiety Depression Insomnia OCD Music.effects
1            Sometimes       2          2        5   1       Improve
2               Rarely       6          7        5   4       Improve
3            Sometimes       7          3        1   5       Improve
4      Very frequently       7          0        0   2       Improve
5            Sometimes       7          3        1   1       Improve
6            Sometimes       5          2        5   1       Improve
7      Very frequently       5          4        8   3     No effect
8            Sometimes       1          1        3   1       Improve
9      Very frequently       9         10        3   0     No effect
10           Sometimes       6          8        1   6       Improve
11           Sometimes       4          2        1   1     No effect
12              Rarely       8          8        3   0       Improve
13              Rarely       6          5        6   8       Improve
14     Very frequently       7          6        6   1     No effect
15              Rarely       4          4        2   1       Improve
16               Never       2          2        2   1     No effect
17           Sometimes      10          6        0   4       Improve
18     Very frequently       6          3        1   1       Improve
19           Sometimes       5          7        7   2       Improve
20     Very frequently       9          9        6   9     No effect
21     Very frequently       6          4        8   8       Improve
22     Very frequently       8          4        3   3        Worsen
23               Never       6          7        8   2       Improve
24               Never       6          5       10   2     No effect
25               Never       8          4        3   0     No effect
26               Never       9          7        5   5       Improve
27     Very frequently       5          3        0   0       Improve
28           Sometimes       8          9        6   4       Improve
29               Never       9          3        7   2       Improve
30              Rarely       2          2        8   0     No effect
31           Sometimes       6          2        6   7     No effect
32           Sometimes       6          1        0   5       Improve
33               Never       2          2        3   2       Improve
34           Sometimes       8          8        9   3       Improve
35              Rarely       5          7        7   0     No effect
36     Very frequently       1          0        7   0       Improve
37     Very frequently       3          2        0   2     No effect
38     Very frequently       8         10       10   0     No effect
39     Very frequently       8          8        5   0       Improve
40               Never       8          5        6   1       Improve
```

41	Never	1	0	0	0	Improve
42	Very frequently	4	5	7	0	No effect
43	Never	4	2	1	0	Improve
44	Sometimes	7	7	0	0	Improve
45	Rarely	1	2	1	0	No effect
46	Very frequently	7	5	7	0	Improve
47	Sometimes	8	7	10	6	Improve
48	Sometimes	7	3	9	2	Improve
49	Never	8	7	4	4	Improve
50	Never	0	0	7	0	Improve
51	Very frequently	4	6	0	2	Improve
52	Rarely	10	8	6	2	No effect
53	Never	6	4	0	0	Improve
54	Never	0	0	0	0	No effect
55	Never	0	0	1	0	No effect
56	Rarely	0	0	0	0	Improve
57	Never	3	3	1	8	No effect
58	Rarely	3	3	0	0	Improve
59	Rarely	3	0	0	5	Improve
60	Never	4	8	9	1	Improve
61	Rarely	10	10	10	9	Improve
62	Never	1	2	2	0	No effect
63	Very frequently	7	10	0	8	Improve
64	Never	10	2	1	10	Worsen
65	Never	2	1	2	2	Improve
66	Rarely	2	0	0	5	No effect
67	Rarely	7	0	0	0	No effect
68	Never	10	5	2	0	Improve
69	Never	6	2	2	0	Improve

	Permissions
1	I understand.
2	I understand.
3	I understand.
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13	I understand.

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14 I understand.
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57 I understand.
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61 I understand.
62 I understand.
63 I understand.
64 I understand.
65 I understand.
66 I understand.
67 I understand.
68 I understand.
69 I understand.
```

```r
# removing the rows where participants stated they do not use a streaming service
music_data_1 <- music_data_1 %>%
  filter(Primary.streaming.service != "I do not use a streaming service.")

music_data_1 %>%
  count(Primary.streaming.service)
```

```
  Primary.streaming.service   n
1               Apple Music  50
2  Other streaming service   49
3                   Pandora  10
4                   Spotify 449
5             YouTube Music  90
```

Modeling for Research Question 1

```r
# Question 1: How does the number of hours spent listening to music per day vary by depressio

# fit our linear regression model
model_rq1_raw <- lm(
  Hours.per.day ~ Depression * Primary.streaming.service +
    Age +
    While.working +
    Instrumentalist +
    Composer +
```

```r
    Exploratory +
    Foreign.languages +
    Anxiety +
    Insomnia +
    OCD +
    Fav.genre +
    Music.effects +
    freq_Classical +
    freq_Country +
    freq_EDM +
    freq_Folk +
    freq_Gospel +
    freq_Hiphop +
    freq_Jazz +
    freq_Kpop +
    freq_Latin +
    freq_Lofi +
    freq_Metal +
    freq_Pop +
    freq_RB +
    freq_Rap +
    freq_Rock +
    freq_Videogamemusic,
  data = music_data_1
)

summary(model_rq1_raw)
```

```
Call:
lm(formula = Hours.per.day ~ Depression * Primary.streaming.service +
    Age + While.working + Instrumentalist + Composer + Exploratory +
    Foreign.languages + Anxiety + Insomnia + OCD + Fav.genre +
    Music.effects + freq_Classical + freq_Country + freq_EDM +
    freq_Folk + freq_Gospel + freq_Hiphop + freq_Jazz + freq_Kpop +
    freq_Latin + freq_Lofi + freq_Metal + freq_Pop + freq_RB +
    freq_Rap + freq_Rock + freq_Videogamemusic, data = music_data_1)

Residuals:
    Min      1Q  Median      3Q     Max
-5.8205 -1.4459 -0.3993  1.0608 17.8079
```

```
Coefficients:
                                                      Estimate
(Intercept)                                          -0.150328
Depression                                            0.399122
Primary.streaming.serviceOther streaming service      2.155139
Primary.streaming.servicePandora                      1.987731
Primary.streaming.serviceSpotify                      2.323886
Primary.streaming.serviceYouTube Music                2.824708
Age                                                  -0.027736
While.workingYes                                      1.711681
InstrumentalistYes                                   -0.942138
ComposerYes                                           0.501449
ExploratoryYes                                        0.208476
Foreign.languagesYes                                  0.011670
Anxiety                                               0.011383
Insomnia                                              0.122338
OCD                                                   0.073755
Fav.genreCountry                                     -0.261941
Fav.genreEDM                                          1.402734
Fav.genreFolk                                        -0.962157
Fav.genreGospel                                      -0.728676
Fav.genreHip hop                                      0.180676
Fav.genreJazz                                         1.192366
Fav.genreK pop                                       -0.559160
Fav.genreLatin                                        3.797748
Fav.genreLofi                                        -0.615007
Fav.genreMetal                                        0.016637
Fav.genrePop                                         -0.429185
Fav.genreR&B                                          0.254203
Fav.genreRap                                          0.789304
Fav.genreRock                                         0.243404
Fav.genreVideo game music                            -0.578321
Music.effectsNo effect                                0.105559
Music.effectsWorsen                                  -0.268246
freq_ClassicalRarely                                 -0.309188
freq_ClassicalSometimes                               0.063121
freq_ClassicalVery frequently                        -0.258064
freq_CountryRarely                                   -0.206280
freq_CountrySometimes                                 0.309509
freq_CountryVery frequently                           0.640736
freq_EDMRarely                                        0.003213
freq_EDMSometimes                                     0.226010
freq_EDMVery frequently                              -0.092925
```

```
freq_FolkRarely                                                     0.081552
freq_FolkSometimes                                                 -0.239450
freq_FolkVery frequently                                            0.850294
freq_GospelRarely                                                   0.070049
freq_GospelSometimes                                               -0.303222
freq_GospelVery frequently                                          1.248700
freq_HiphopRarely                                                  -0.592983
freq_HiphopSometimes                                               -0.271432
freq_HiphopVery frequently                                         -0.615644
freq_JazzRarely                                                     0.130704
freq_JazzSometimes                                                  0.602272
freq_JazzVery frequently                                            0.398442
freq_KpopRarely                                                     0.413204
freq_KpopSometimes                                                  0.384729
freq_KpopVery frequently                                            0.249966
freq_LatinRarely                                                    0.535220
freq_LatinSometimes                                                 0.071835
freq_LatinVery frequently                                           0.340177
freq_LofiRarely                                                    -0.289755
freq_LofiSometimes                                                 -0.163425
freq_LofiVery frequently                                            0.434449
freq_MetalRarely                                                   -0.031746
freq_MetalSometimes                                                 0.468273
freq_MetalVery frequently                                           0.162659
freq_PopRarely                                                      0.168971
freq_PopSometimes                                                  -0.912145
freq_PopVery frequently                                            -0.830877
freq_RBRarely                                                      -0.144401
freq_RBSometimes                                                    0.670428
freq_RBVery frequently                                             -0.091912
freq_RapRarely                                                      0.315587
freq_RapSometimes                                                   0.058571
freq_RapVery frequently                                             0.932615
freq_RockRarely                                                     0.477393
freq_RockSometimes                                                  0.109539
freq_RockVery frequently                                            0.345797
freq_VideogamemusicRarely                                          -0.014055
freq_VideogamemusicSometimes                                        0.304364
freq_VideogamemusicVery frequently                                 -0.748022
Depression:Primary.streaming.serviceOther streaming service -0.585820
Depression:Primary.streaming.servicePandora                        -0.584946
Depression:Primary.streaming.serviceSpotify                        -0.409516
Depression:Primary.streaming.serviceYouTube Music                  -0.616925
```

	Std. Error	t value
(Intercept)	1.230860	-0.122
Depression	0.139433	2.862
Primary.streaming.serviceOther streaming service	1.063987	2.026
Primary.streaming.servicePandora	1.566865	1.269
Primary.streaming.serviceSpotify	0.859829	2.703
Primary.streaming.serviceYouTube Music	0.940640	3.003
Age	0.012952	-2.141
While.workingYes	0.287702	5.949
InstrumentalistYes	0.284173	-3.315
ComposerYes	0.330255	1.518
ExploratoryYes	0.277352	0.752
Foreign.languagesYes	0.252383	0.046
Anxiety	0.050293	0.226
Insomnia	0.040279	3.037
OCD	0.041647	1.771
Fav.genreCountry	1.045464	-0.251
Fav.genreEDM	0.853106	1.644
Fav.genreFolk	0.861645	-1.117
Fav.genreGospel	1.776859	-0.410
Fav.genreHip hop	0.845239	0.214
Fav.genreJazz	0.931780	1.280
Fav.genreK pop	0.949612	-0.589
Fav.genreLatin	2.083727	1.823
Fav.genreLofi	1.100204	-0.559
Fav.genreMetal	0.775253	0.021
Fav.genrePop	0.678593	-0.632
Fav.genreR&B	0.821423	0.309
Fav.genreRap	0.895099	0.882
Fav.genreRock	0.656264	0.371
Fav.genreVideo game music	0.810298	-0.714
Music.effectsNo effect	0.276362	0.382
Music.effectsWorsen	0.749394	-0.358
freq_ClassicalRarely	0.308653	-1.002
freq_ClassicalSometimes	0.337484	0.187
freq_ClassicalVery frequently	0.488864	-0.528
freq_CountryRarely	0.273610	-0.754
freq_CountrySometimes	0.361029	0.857
freq_CountryVery frequently	0.665983	0.962
freq_EDMRarely	0.292896	0.011
freq_EDMSometimes	0.329195	0.687
freq_EDMVery frequently	0.474976	-0.196
freq_FolkRarely	0.293811	0.278

freq_FolkSometimes	0.353867	-0.677
freq_FolkVery frequently	0.459447	1.851
freq_GospelRarely	0.312318	0.224
freq_GospelSometimes	0.445614	-0.680
freq_GospelVery frequently	1.118153	1.117
freq_HiphopRarely	0.378722	-1.566
freq_HiphopSometimes	0.448260	-0.606
freq_HiphopVery frequently	0.590202	-1.043
freq_JazzRarely	0.281594	0.464
freq_JazzSometimes	0.338712	1.778
freq_JazzVery frequently	0.573379	0.695
freq_KpopRarely	0.295424	1.399
freq_KpopSometimes	0.418258	0.920
freq_KpopVery frequently	0.492109	0.508
freq_LatinRarely	0.302947	1.767
freq_LatinSometimes	0.375263	0.191
freq_LatinVery frequently	0.602392	0.565
freq_LofiRarely	0.292985	-0.989
freq_LofiSometimes	0.325718	-0.502
freq_LofiVery frequently	0.446224	0.974
freq_MetalRarely	0.315762	-0.101
freq_MetalSometimes	0.368603	1.270
freq_MetalVery frequently	0.473578	0.343
freq_PopRarely	0.511654	0.330
freq_PopSometimes	0.502195	-1.816
freq_PopVery frequently	0.531905	-1.562
freq_RBRarely	0.323272	-0.447
freq_RBSometimes	0.359720	1.864
freq_RBVery frequently	0.470120	-0.196
freq_RapRarely	0.378824	0.833
freq_RapSometimes	0.434448	0.135
freq_RapVery frequently	0.570699	1.634
freq_RockRarely	0.456858	1.045
freq_RockSometimes	0.427629	0.256
freq_RockVery frequently	0.462190	0.748
freq_VideogamemusicRarely	0.298573	-0.047
freq_VideogamemusicSometimes	0.325813	0.934
freq_VideogamemusicVery frequently	0.427114	-1.751
Depression:Primary.streaming.serviceOther streaming service	0.186165	-3.147
Depression:Primary.streaming.servicePandora	0.290771	-2.012
Depression:Primary.streaming.serviceSpotify	0.144633	-2.831
Depression:Primary.streaming.serviceYouTube Music	0.165277	-3.733

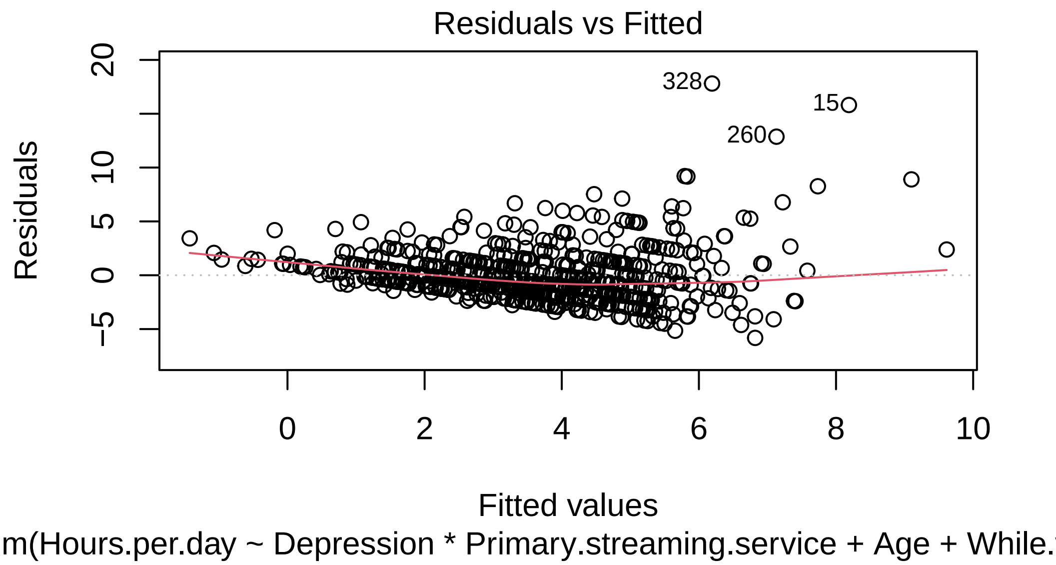
Pr(>|t|)

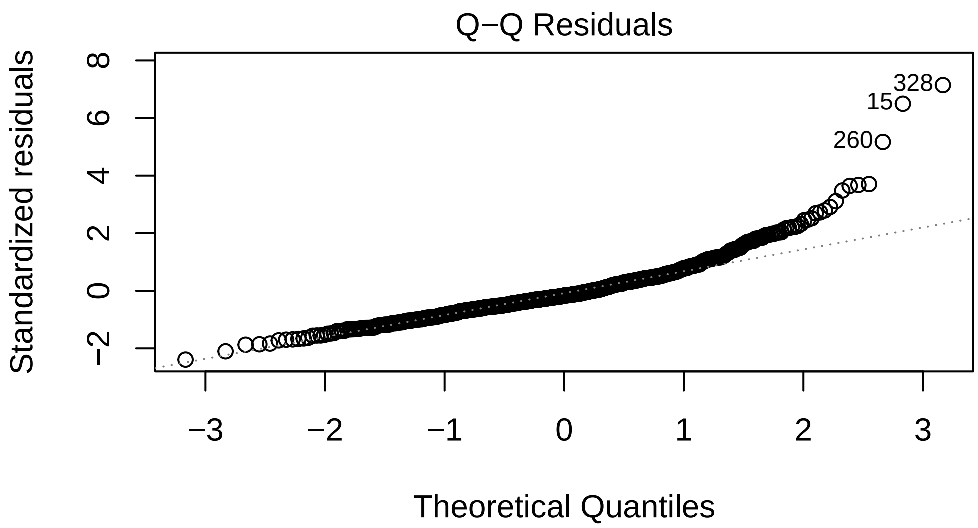
```
(Intercept)                                         0.902838
Depression                                          0.004360 **
Primary.streaming.serviceOther streaming service    0.043283 *
Primary.streaming.servicePandora                    0.205106
Primary.streaming.serviceSpotify                    0.007084 **
Primary.streaming.serviceYouTube Music              0.002792 **
Age                                                 0.032664 *
While.workingYes                                    4.72e-09 ***
InstrumentalistYes                                  0.000974 ***
ComposerYes                                         0.129481
ExploratoryYes                                      0.452566
Foreign.languagesYes                                0.963137
Anxiety                                             0.821024
Insomnia                                            0.002498 **
OCD                                                 0.077110 .
Fav.genreCountry                                    0.802253
Fav.genreEDM                                        0.100678
Fav.genreFolk                                       0.264619
Fav.genreGospel                                     0.681894
Fav.genreHip hop                                    0.830813
Fav.genreJazz                                       0.201189
Fav.genreK pop                                      0.556211
Fav.genreLatin                                      0.068897 .
Fav.genreLofi                                       0.576388
Fav.genreMetal                                      0.982886
Fav.genrePop                                        0.527341
Fav.genreR&B                                        0.757080
Fav.genreRap                                        0.378257
Fav.genreRock                                       0.710856
Fav.genreVideo game music                           0.475699
Music.effectsNo effect                              0.702636
Music.effectsWorsen                                 0.720514
freq_ClassicalRarely                                0.316902
freq_ClassicalSometimes                             0.851702
freq_ClassicalVery frequently                       0.597787
freq_CountryRarely                                  0.451211
freq_CountrySometimes                               0.391646
freq_CountryVery frequently                         0.336417
freq_EDMRarely                                      0.991251
freq_EDMSometimes                                   0.492645
freq_EDMVery frequently                             0.844961
freq_FolkRarely                                     0.781447
freq_FolkSometimes                                  0.498894
```

```
freq_FolkVery frequently                                  0.064737 .
freq_GospelRarely                                         0.822615
freq_GospelSometimes                                      0.496493
freq_GospelVery frequently                                0.264576
freq_HiphopRarely                                         0.117969
freq_HiphopSometimes                                      0.545074
freq_HiphopVery frequently                                0.297345
freq_JazzRarely                                           0.642714
freq_JazzSometimes                                        0.075923 .
freq_JazzVery frequently                                  0.487404
freq_KpopRarely                                           0.162458
freq_KpopSometimes                                        0.358052
freq_KpopVery frequently                                  0.611688
freq_LatinRarely                                          0.077817 .
freq_LatinSometimes                                       0.848261
freq_LatinVery frequently                                 0.572495
freq_LofiRarely                                           0.323098
freq_LofiSometimes                                        0.616048
freq_LofiVery frequently                                  0.330667
freq_MetalRarely                                          0.919953
freq_MetalSometimes                                       0.204466
freq_MetalVery frequently                                 0.731374
freq_PopRarely                                            0.741337
freq_PopSometimes                                         0.069852 .
freq_PopVery frequently                                   0.118831
freq_RBRarely                                             0.655273
freq_RBSometimes                                          0.062877 .
freq_RBVery frequently                                    0.845067
freq_RapRarely                                            0.405159
freq_RapSometimes                                         0.892805
freq_RapVery frequently                                   0.102782
freq_RockRarely                                           0.296494
freq_RockSometimes                                        0.797925
freq_RockVery frequently                                  0.454669
freq_VideogamemusicRarely                                 0.962472
freq_VideogamemusicSometimes                              0.350617
freq_VideogamemusicVery frequently                        0.080431 .
Depression:Primary.streaming.serviceOther streaming service 0.001738 **
Depression:Primary.streaming.servicePandora               0.044726 *
Depression:Primary.streaming.serviceSpotify               0.004800 **
Depression:Primary.streaming.serviceYouTube Music         0.000209 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

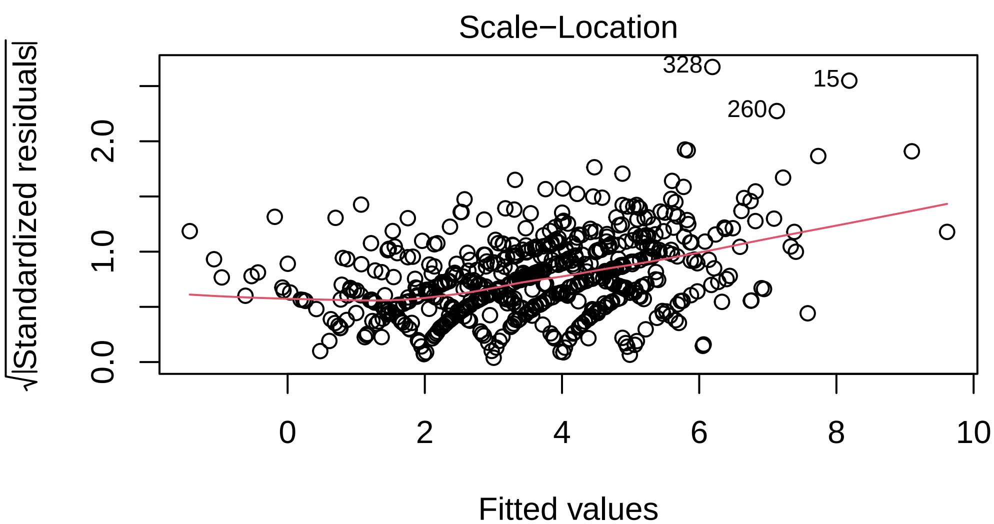
```
Residual standard error: 2.633 on 564 degrees of freedom
Multiple R-squared:  0.2972,	Adjusted R-squared:  0.1938
F-statistic: 2.874 on 83 and 564 DF,  p-value: 2.465e-13
```

```r
plot(model_rq1_raw)
```

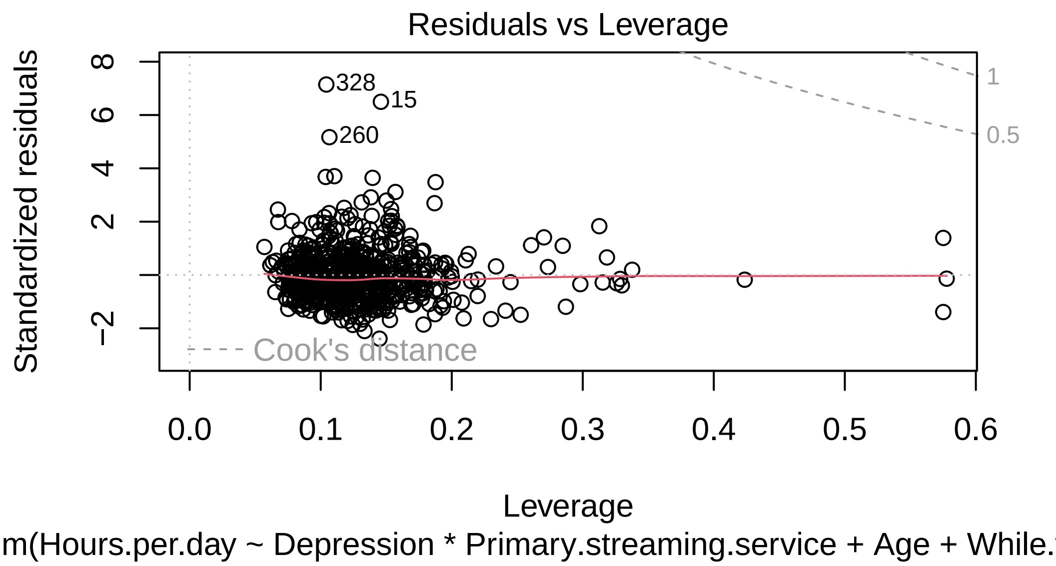




m(Hours.per.day ~ Depression * Primary.streaming.service + Age + While.



m(Hours.per.day ~ Depression * Primary.streaming.service + Age + While.




```r
# checking for multicollinearity
vif(model_rq1_raw, type = "predictor")
```

```
GVIFs computed for predictors

                                GVIF Df GVIF^(1/(2*Df))
Depression                  5.578225  9        1.100201
Primary.streaming.service   5.578225  9        1.100201
Age                         1.894314  1        1.376341
While.working               1.240283  1        1.113680
Instrumentalist             1.631863  1        1.277444
Composer                    1.415396  1        1.189704
Exploratory                 1.330852  1        1.153626
Foreign.languages           1.464064  1        1.209985
Anxiety                     1.801603  1        1.342238
Insomnia                    1.416839  1        1.190311
OCD                         1.314957  1        1.146716
Fav.genre                2413.101621 15        1.296440
Music.effects               1.431300  2        1.093787
freq_Classical              3.145646  3        1.210463
freq_Country                4.221584  3        1.271294
freq_EDM                    3.209085  3        1.214498
```

```
freq_Folk                  2.777919  3        1.185641
freq_Gospel                3.225607  3        1.215538
freq_Hiphop                8.044445  3        1.415520
freq_Jazz                  3.086056  3        1.206611
freq_Kpop                  3.264117  3        1.217945
freq_Latin                 2.518822  3        1.166450
freq_Lofi                  2.412175  3        1.158070
freq_Metal                 4.752521  3        1.296644
freq_Pop                   2.685629  3        1.178983
freq_RB                    3.839185  3        1.251334
freq_Rap                   8.130319  3        1.418027
freq_Rock                  3.488711  3        1.231528
freq_Videogamemusic        2.535561  3        1.167739
                                   Interacts With
Depression                Primary.streaming.service
Primary.streaming.service                Depression
Age                                            --
While.working                                  --
Instrumentalist                                --
Composer                                       --
Exploratory                                    --
Foreign.languages                              --
Anxiety                                        --
Insomnia                                       --
OCD                                            --
Fav.genre                                      --
Music.effects                                  --
freq_Classical                                 --
freq_Country                                   --
freq_EDM                                       --
freq_Folk                                      --
freq_Gospel                                    --
freq_Hiphop                                    --
freq_Jazz                                      --
freq_Kpop                                      --
freq_Latin                                     --
freq_Lofi                                      --
freq_Metal                                     --
freq_Pop                                       --
freq_RB                                        --
freq_Rap                                       --
freq_Rock                                      --
freq_Videogamemusic                            --
```

```
Depression                                                  Age, While.working, Instrumentali
Primary.streaming.service                                   Age, While.working, Instrumentali
Age                        Depression, Primary.streaming.service, While.working, Instrumentali
While.working                        Depression, Primary.streaming.service, Age, Instrumentali
Instrumentalist                        Depression, Primary.streaming.service, Age, While.worki
Composer                        Depression, Primary.streaming.service, Age, While.working, Ins
Exploratory                        Depression, Primary.streaming.service, Age, While.working,
Foreign.languages                        Depression, Primary.streaming.service, Age, While.wo
Anxiety                        Depression, Primary.streaming.service, Age, While.working, Inst
Insomnia                        Depression, Primary.streaming.service, Age, While.working, Ins
OCD                        Depression, Primary.streaming.service, Age, While.working, Instrume
Fav.genre                        Depression, Primary.streaming.service, Age, While.working, In
Music.effects                        Depression, Primary.streaming.service, Age, While.working
freq_Classical                        Depression, Primary.streaming.service, Age, While.workin
freq_Country                        Depression, Primary.streaming.service, Age, While.working
freq_EDM                        Depression, Primary.streaming.service, Age, While.working, Ins
freq_Folk                        Depression, Primary.streaming.service, Age, While.working, In
freq_Gospel                        Depression, Primary.streaming.service, Age, While.working,
freq_Hiphop                        Depression, Primary.streaming.service, Age, While.working,
freq_Jazz                        Depression, Primary.streaming.service, Age, While.working, In
freq_Kpop                        Depression, Primary.streaming.service, Age, While.working, In
freq_Latin                        Depression, Primary.streaming.service, Age, While.working, I
freq_Lofi                        Depression, Primary.streaming.service, Age, While.working, In
freq_Metal                        Depression, Primary.streaming.service, Age, While.working, I
freq_Pop                        Depression, Primary.streaming.service, Age, While.working, Ins
freq_RB                        Depression, Primary.streaming.service, Age, While.working, Inst
freq_Rap                        Depression, Primary.streaming.service, Age, While.working, Ins
freq_Rock                        Depression, Primary.streaming.service, Age, While.working, In
freq_Videogamemusic                        Depression, Primary.streaming.service, Age, While.w
```

Model with Intentional/Relevant Predictors RQ1

```r
# fit our linear regression model
model_rq1 <- lm(
  Hours.per.day ~ Depression * Primary.streaming.service +
    Age +
    While.working +
    Instrumentalist +
    Composer +
```

```r
    Exploratory +
    Foreign.languages +
    Anxiety +
    Insomnia +
    OCD +
    Fav.genre +
    Music.effects,
  data = music_data_1
)

summary(model_rq1)
```

```

Call:
lm(formula = Hours.per.day ~ Depression * Primary.streaming.service +
    Age + While.working + Instrumentalist + Composer + Exploratory +
    Foreign.languages + Anxiety + Insomnia + OCD + Fav.genre +
    Music.effects, data = music_data_1)

Residuals:
   Min     1Q Median     3Q    Max
-4.540 -1.662 -0.512  0.993 18.913

Coefficients:
                                                         Estimate Std. Error
(Intercept)                                              -0.86118    1.01955
Depression                                                0.45316    0.13858
Primary.streaming.serviceOther streaming service          2.39086    1.04135
Primary.streaming.servicePandora                          2.49022    1.52774
Primary.streaming.serviceSpotify                          2.30355    0.84945
Primary.streaming.serviceYouTube Music                    2.83507    0.92889
Age                                                      -0.01682    0.01126
While.workingYes                                          1.83687    0.27687
InstrumentalistYes                                       -0.84526    0.26712
ComposerYes                                               0.57879    0.32447
ExploratoryYes                                            0.40967    0.26251
Foreign.languagesYes                                      0.19630    0.22992
Anxiety                                                  -0.01158    0.04869
Insomnia                                                  0.12078    0.03940
OCD                                                       0.07333    0.04099
Fav.genreCountry                                          0.44927    0.72251
Fav.genreEDM                                              1.32603    0.64356
```

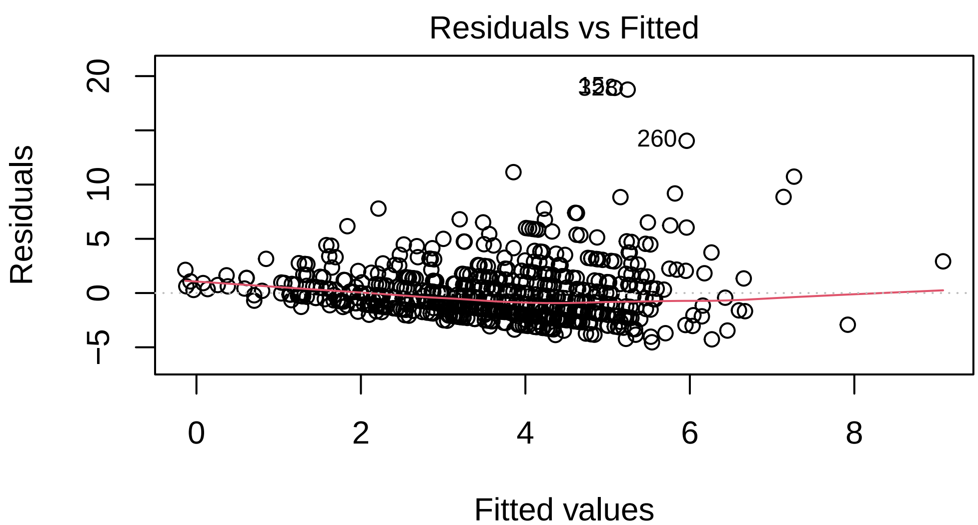
```
Fav.genreFolk                                                   0.02657    0.68518
Fav.genreGospel                                                 0.54158    1.36426
Fav.genreHip hop                                                0.47652    0.65632
Fav.genreJazz                                                   1.92971    0.75268
Fav.genreK pop                                                 -0.32687    0.75575
Fav.genreLatin                                                  4.88370    1.96889
Fav.genreLofi                                                  -0.19278    0.97904
Fav.genreMetal                                                  0.18570    0.54055
Fav.genrePop                                                   -0.47212    0.51839
Fav.genreR&B                                                    0.11123    0.66038
Fav.genreRap                                                    1.44534    0.75606
Fav.genreRock                                                   0.50531    0.49346
Fav.genreVideo game music                                      -0.74626    0.66918
Music.effectsNo effect                                          0.12482    0.26919
Music.effectsWorsen                                            -0.52436    0.73095
Depression:Primary.streaming.serviceOther streaming service -0.60737    0.18395
Depression:Primary.streaming.servicePandora                    -0.69263    0.28670
Depression:Primary.streaming.serviceSpotify                    -0.43332    0.14372
Depression:Primary.streaming.serviceYouTube Music              -0.62748    0.16495
                                                               t value Pr(>|t|)
(Intercept)                                                     -0.845 0.398628
Depression                                                       3.270 0.001135
Primary.streaming.serviceOther streaming service                 2.296 0.022018
Primary.streaming.servicePandora                                 1.630 0.103616
Primary.streaming.serviceSpotify                                 2.712 0.006879
Primary.streaming.serviceYouTube Music                           3.052 0.002371
Age                                                             -1.493 0.135839
While.workingYes                                                 6.634 7.18e-11
InstrumentalistYes                                              -3.164 0.001632
ComposerYes                                                      1.784 0.074947
ExploratoryYes                                                   1.561 0.119142
Foreign.languagesYes                                             0.854 0.393549
Anxiety                                                         -0.238 0.812085
Insomnia                                                         3.066 0.002268
OCD                                                              1.789 0.074132
Fav.genreCountry                                                 0.622 0.534293
Fav.genreEDM                                                     2.060 0.039777
Fav.genreFolk                                                    0.039 0.969082
Fav.genreGospel                                                  0.397 0.691525
Fav.genreHip hop                                                 0.726 0.468086
Fav.genreJazz                                                    2.564 0.010591
Fav.genreK pop                                                  -0.433 0.665528
Fav.genreLatin                                                   2.480 0.013390
```

```
Fav.genreLofi                                                      -0.197 0.843967
Fav.genreMetal                                                      0.344 0.731313
Fav.genrePop                                                       -0.911 0.362784
Fav.genreR&B                                                        0.168 0.866296
Fav.genreRap                                                        1.912 0.056384
Fav.genreRock                                                       1.024 0.306238
Fav.genreVideo game music                                          -1.115 0.265208
Music.effectsNo effect                                              0.464 0.643027
Music.effectsWorsen                                                -0.717 0.473425
Depression:Primary.streaming.serviceOther streaming service  -3.302 0.001017
Depression:Primary.streaming.servicePandora                        -2.416 0.015990
Depression:Primary.streaming.serviceSpotify                        -3.015 0.002677
Depression:Primary.streaming.serviceYouTube Music                  -3.804 0.000157

(Intercept)
Depression                                                    **
Primary.streaming.serviceOther streaming service              *
Primary.streaming.servicePandora
Primary.streaming.serviceSpotify                              **
Primary.streaming.serviceYouTube Music                        **
Age
While.workingYes                                              ***
InstrumentalistYes                                            **
ComposerYes                                                   .
ExploratoryYes
Foreign.languagesYes
Anxiety
Insomnia                                                      **
OCD                                                           .
Fav.genreCountry
Fav.genreEDM                                                  *
Fav.genreFolk
Fav.genreGospel
Fav.genreHip hop
Fav.genreJazz                                                 *
Fav.genreK pop
Fav.genreLatin                                                *
Fav.genreLofi
Fav.genreMetal
Fav.genrePop
Fav.genreR&B
Fav.genreRap                                                  .
Fav.genreRock
```

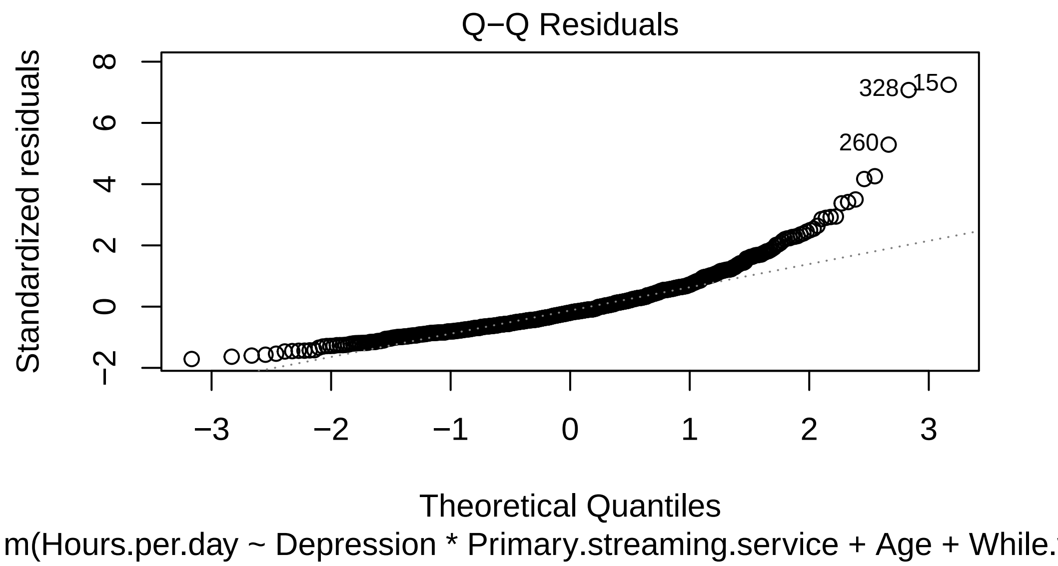
```
Fav.genreVideo game music
Music.effectsNo effect
Music.effectsWorsen
Depression:Primary.streaming.serviceOther streaming service **
Depression:Primary.streaming.servicePandora                 *
Depression:Primary.streaming.serviceSpotify                 **
Depression:Primary.streaming.serviceYouTube Music           ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.691 on 612 degrees of freedom
Multiple R-squared:  0.2035,    Adjusted R-squared:  0.158
F-statistic: 4.469 on 35 and 612 DF,  p-value: 5.388e-15
```

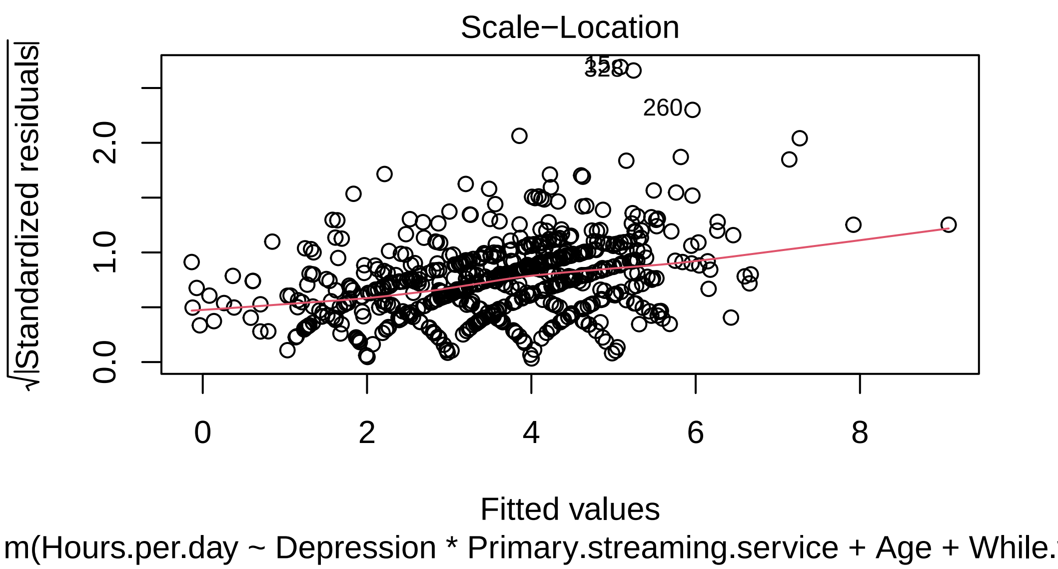
Diagnostics for Research Question 1 Model

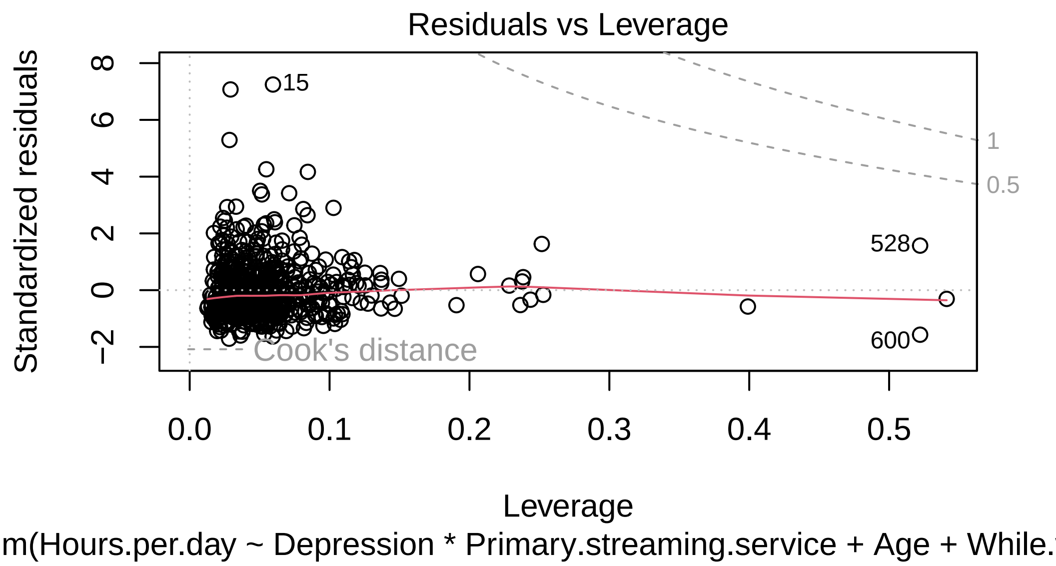
```r
plot(model_rq1)
```



m(Hours.per.day ~ Depression * Primary.streaming.service + Age + While.


Q−Q Residuals
Standardized residuals
Theoretical Quantiles
m(Hours.per.day ~ Depression * Primary.streaming.service + Age + While.
328
15
260


Scale−Location
√|Standardized residuals|
Fitted values
m(Hours.per.day ~ Depression * Primary.streaming.service + Age + While.
15
328
260

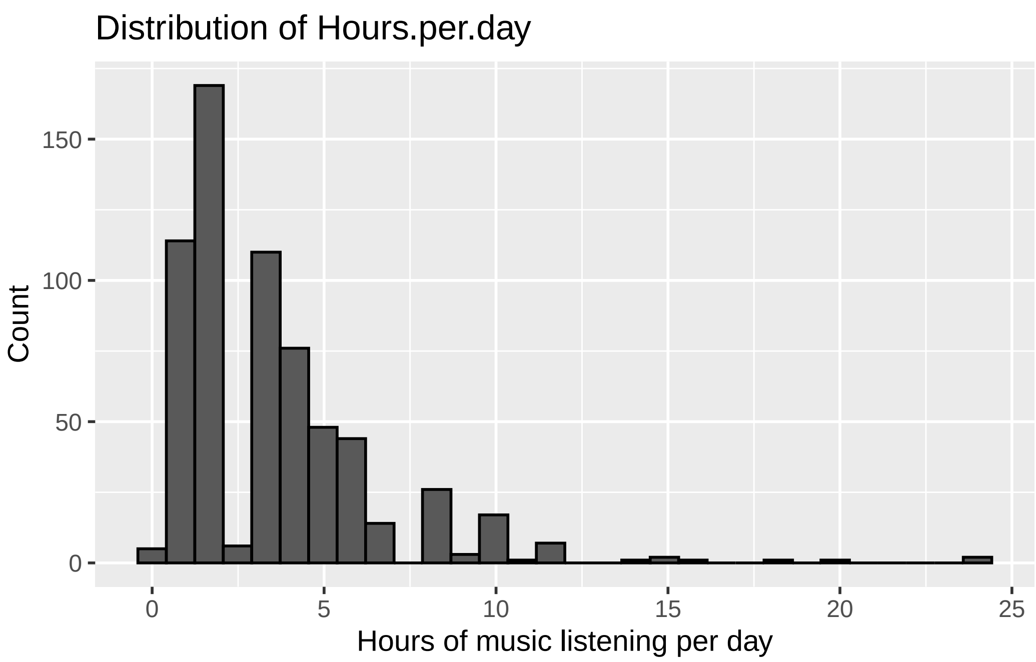



```r
# checking for multicollinearity
vif(model_rq1, type = "predictor")
```

```
GVIFs computed for predictors

                            GVIF Df GVIF^(1/(2*Df))         Interacts With
Depression              2.602031  9        1.054564 Primary.streaming.service
Primary.streaming.service 2.602031  9      1.054564              Depression
Age                     1.371260  1        1.171008                      --
While.working           1.099809  1        1.048718                      --
Instrumentalist         1.380580  1        1.174981                      --
Composer                1.308104  1        1.143724                      --
Exploratory             1.141526  1        1.068422                      --
Foreign.languages       1.163337  1        1.078581                      --
Anxiety                 1.617052  1        1.271633                      --
Insomnia                1.297880  1        1.139246                      --
OCD                     1.219763  1        1.104429                      --
Fav.genre               2.337967 15        1.028714                      --
Music.effects           1.183207  2        1.042954                      --

Depression                                           Age, While.working, Instrumentali
Primary.streaming.service                            Age, While.working, Instrumentali
```

```
Age                          Depression, Primary.streaming.service, While.working, Instrumentali
While.working                          Depression, Primary.streaming.service, Age, Instrumentali
Instrumentalist                          Depression, Primary.streaming.service, Age, While.worki
Composer                          Depression, Primary.streaming.service, Age, While.working, Ins
Exploratory                          Depression, Primary.streaming.service, Age, While.working,
Foreign.languages                          Depression, Primary.streaming.service, Age, While.wo
Anxiety                          Depression, Primary.streaming.service, Age, While.working, Inst
Insomnia                          Depression, Primary.streaming.service, Age, While.working, Ins
OCD                          Depression, Primary.streaming.service, Age, While.working, Instrume
Fav.genre                          Depression, Primary.streaming.service, Age, While.working, In
Music.effects                          Depression, Primary.streaming.service, Age, While.workin
```

```
p1 <- ggplot(music_data_1, aes(x = Hours.per.day)) +
geom_histogram(bins = 30, color = "black") +
labs(
x = "Hours of music listening per day",
y = "Count",
title = "Distribution of Hours.per.day"
)

p1
```



Log Transformation of Outcome Variable RQ1

```
# build log-transformed outcome
music_data_1 <- music_data_1 %>%
  mutate(log_hours = log(Hours.per.day + 0.1))

# run log-transform
model_rq1_log <- lm(
  log_hours ~ Depression * Primary.streaming.service +
    Age +
    While.working +
    Instrumentalist +
    Composer +
    Exploratory +
    Foreign.languages +
    Anxiety +
    Insomnia +
    OCD +
    Fav.genre +
    Music.effects,
  data = music_data_1
)

summary(model_rq1_log)
```

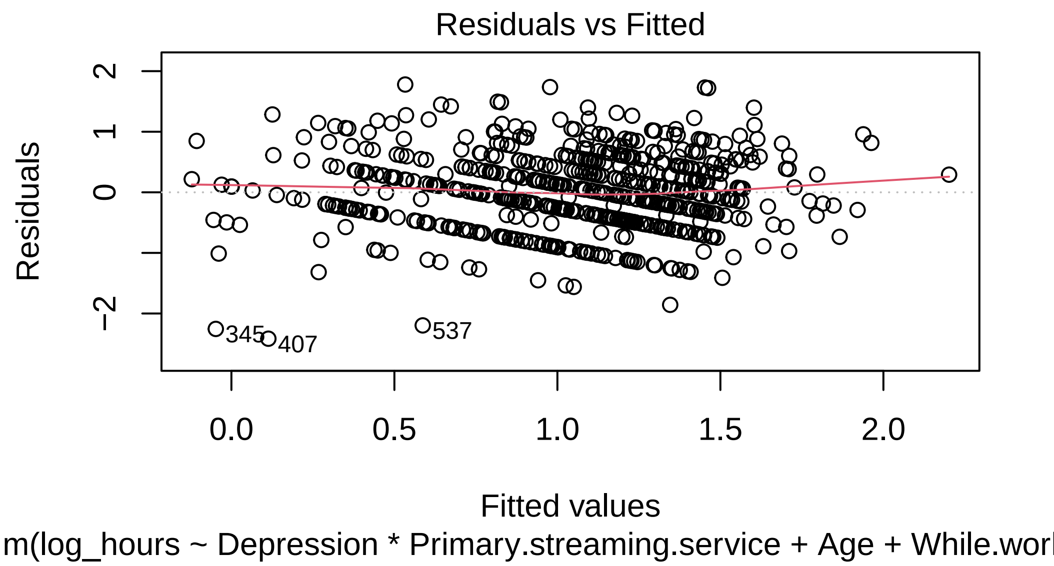
```
Call:
lm(formula = log_hours ~ Depression * Primary.streaming.service +
    Age + While.working + Instrumentalist + Composer + Exploratory +
    Foreign.languages + Anxiety + Insomnia + OCD + Fav.genre +
    Music.effects, data = music_data_1)

Residuals:
     Min       1Q   Median       3Q      Max
-2.41602 -0.41604 -0.00668  0.42421  1.77931

Coefficients:
                                                            Estimate
(Intercept)                                                -0.146165
Depression                                                  0.109401
Primary.streaming.serviceOther streaming service            0.522094
```

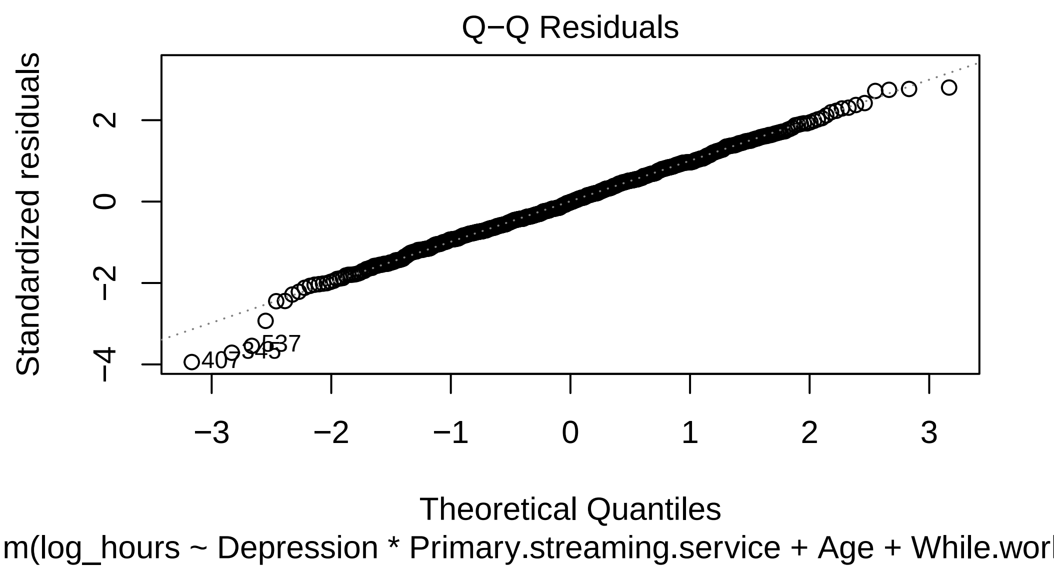
```
Primary.streaming.servicePandora                              0.749074
Primary.streaming.serviceSpotify                              0.588674
Primary.streaming.serviceYouTube Music                        0.633878
Age                                                          -0.004860
While.workingYes                                              0.631290
InstrumentalistYes                                           -0.204095
ComposerYes                                                   0.171280
ExploratoryYes                                                0.114866
Foreign.languagesYes                                          0.030338
Anxiety                                                      -0.002759
Insomnia                                                      0.025883
OCD                                                           0.022209
Fav.genreCountry                                              0.125853
Fav.genreEDM                                                  0.223948
Fav.genreFolk                                                -0.020408
Fav.genreGospel                                               0.057613
Fav.genreHip hop                                              0.098355
Fav.genreJazz                                                 0.396245
Fav.genreK pop                                               -0.086284
Fav.genreLatin                                                0.953296
Fav.genreLofi                                                 0.012051
Fav.genreMetal                                                0.109551
Fav.genrePop                                                 -0.131399
Fav.genreR&B                                                  0.115308
Fav.genreRap                                                  0.285495
Fav.genreRock                                                 0.065427
Fav.genreVideo game music                                    -0.332080
Music.effectsNo effect                                       -0.011814
Music.effectsWorsen                                          -0.287325
Depression:Primary.streaming.serviceOther streaming service -0.147964
Depression:Primary.streaming.servicePandora                  -0.206928
Depression:Primary.streaming.serviceSpotify                  -0.104683
Depression:Primary.streaming.serviceYouTube Music            -0.140281
                                                            Std. Error t value
(Intercept)                                                   0.244701  -0.597
Depression                                                    0.033259   3.289
Primary.streaming.serviceOther streaming service              0.249934   2.089
Primary.streaming.servicePandora                              0.366672   2.043
Primary.streaming.serviceSpotify                              0.203875   2.887
Primary.streaming.serviceYouTube Music                        0.222942   2.843
Age                                                           0.002703  -1.798
While.workingYes                                              0.066452   9.500
InstrumentalistYes                                            0.064111  -3.183
```

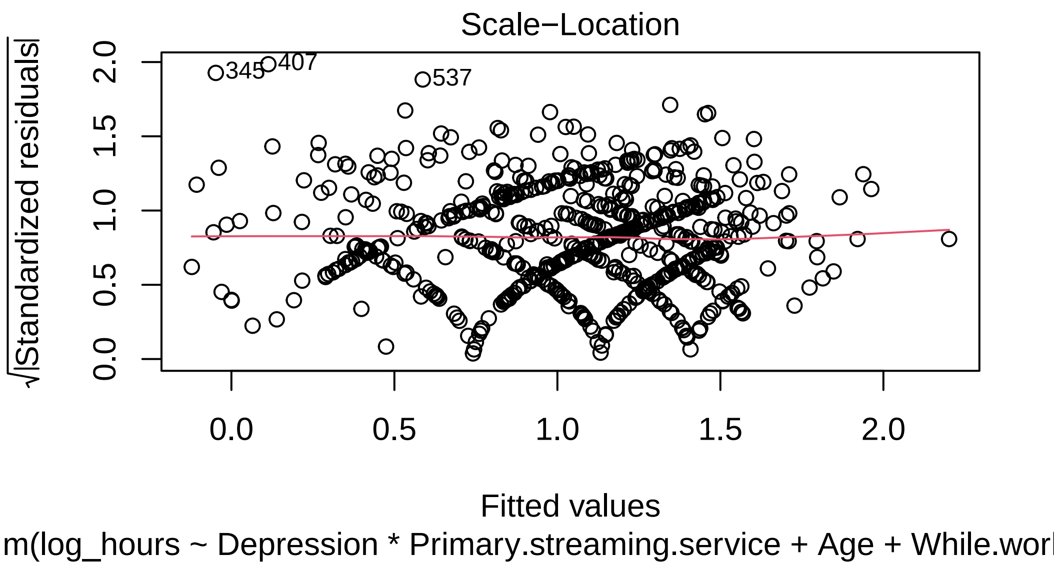
```
ComposerYes                                                 0.077875   2.199
ExploratoryYes                                              0.063005   1.823
Foreign.languagesYes                                        0.055182   0.550
Anxiety                                                     0.011687  -0.236
Insomnia                                                    0.009456   2.737
OCD                                                         0.009839   2.257
Fav.genreCountry                                            0.173409   0.726
Fav.genreEDM                                                0.154461   1.450
Fav.genreFolk                                               0.164449  -0.124
Fav.genreGospel                                             0.327433   0.176
Fav.genreHip hop                                            0.157522   0.624
Fav.genreJazz                                               0.180649   2.193
Fav.genreK pop                                              0.181387  -0.476
Fav.genreLatin                                              0.472550   2.017
Fav.genreLofi                                               0.234978   0.051
Fav.genreMetal                                              0.129736   0.844
Fav.genrePop                                                0.124418  -1.056
Fav.genreR&B                                                0.158498   0.728
Fav.genreRap                                                0.181461   1.573
Fav.genreRock                                               0.118435   0.552
Fav.genreVideo game music                                   0.160608  -2.068
Music.effectsNo effect                                      0.064609  -0.183
Music.effectsWorsen                                         0.175434  -1.638
Depression:Primary.streaming.serviceOther streaming service 0.044150  -3.351
Depression:Primary.streaming.servicePandora                 0.068811  -3.007
Depression:Primary.streaming.serviceSpotify                 0.034495  -3.035
Depression:Primary.streaming.serviceYouTube Music           0.039590  -3.543
                                                           Pr(>|t|)
(Intercept)                                                0.550513
Depression                                                 0.001062 **
Primary.streaming.serviceOther streaming service           0.037127 *
Primary.streaming.servicePandora                           0.041490 *
Primary.streaming.serviceSpotify                           0.004021 **
Primary.streaming.serviceYouTube Music                     0.004615 **
Age                                                        0.072664 .
While.workingYes                                            < 2e-16 ***
InstrumentalistYes                                         0.001529 **
ComposerYes                                                0.028221 *
ExploratoryYes                                             0.068771 .
Foreign.languagesYes                                       0.582672
Anxiety                                                    0.813438
Insomnia                                                   0.006376 **
OCD                                                        0.024336 *
```

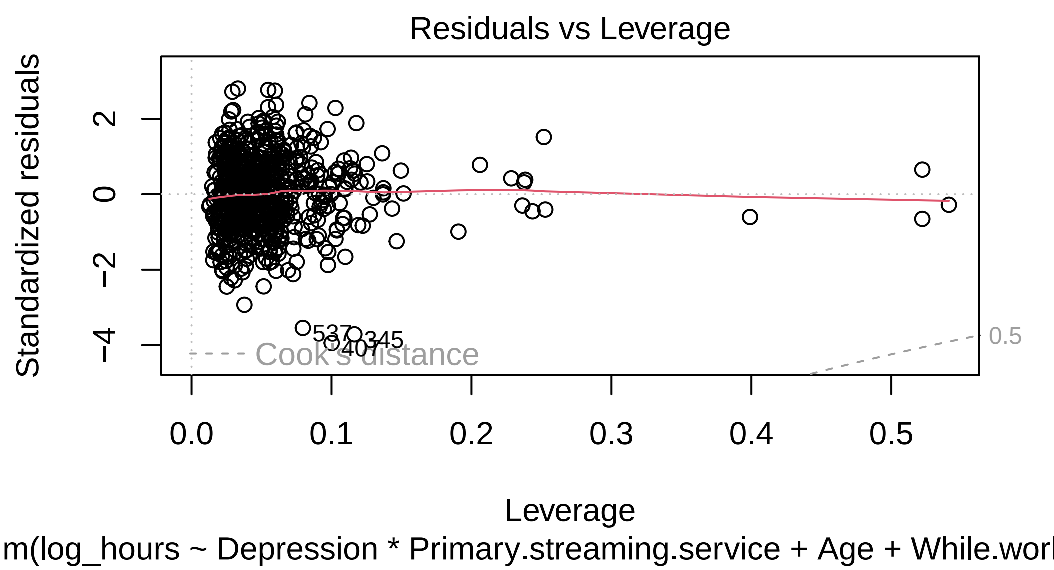
```
Fav.genreCountry                                          0.468266
Fav.genreEDM                                              0.147606
Fav.genreFolk                                             0.901278
Fav.genreGospel                                           0.860389
Fav.genreHip hop                                          0.532603
Fav.genreJazz                                             0.028650 *
Fav.genreK pop                                            0.634465
Fav.genreLatin                                            0.044096 *
Fav.genreLofi                                             0.959115
Fav.genreMetal                                            0.398764
Fav.genrePop                                              0.291336
Fav.genreR&B                                              0.467194
Fav.genreRap                                              0.116163
Fav.genreRock                                             0.580854
Fav.genreVideo game music                                 0.039094 *
Music.effectsNo effect                                    0.854967
Music.effectsWorsen                                       0.101979
Depression:Primary.streaming.serviceOther streaming service 0.000854 ***
Depression:Primary.streaming.servicePandora               0.002745 **
Depression:Primary.streaming.serviceSpotify               0.002510 **
Depression:Primary.streaming.serviceYouTube Music         0.000425 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6459 on 612 degrees of freedom
Multiple R-squared:  0.274, Adjusted R-squared:  0.2325
F-statistic: 6.599 on 35 and 612 DF,  p-value: < 2.2e-16
```

```r
# diagnostic plots for log-transformed model
plot(model_rq1_log)
```


Residuals vs Fitted
Residuals
Fitted values
m(log_hours ~ Depression * Primary.streaming.service + Age + While.work
345
407
537
2
1
0
−2
0.0
0.5
1.0
1.5
2.0

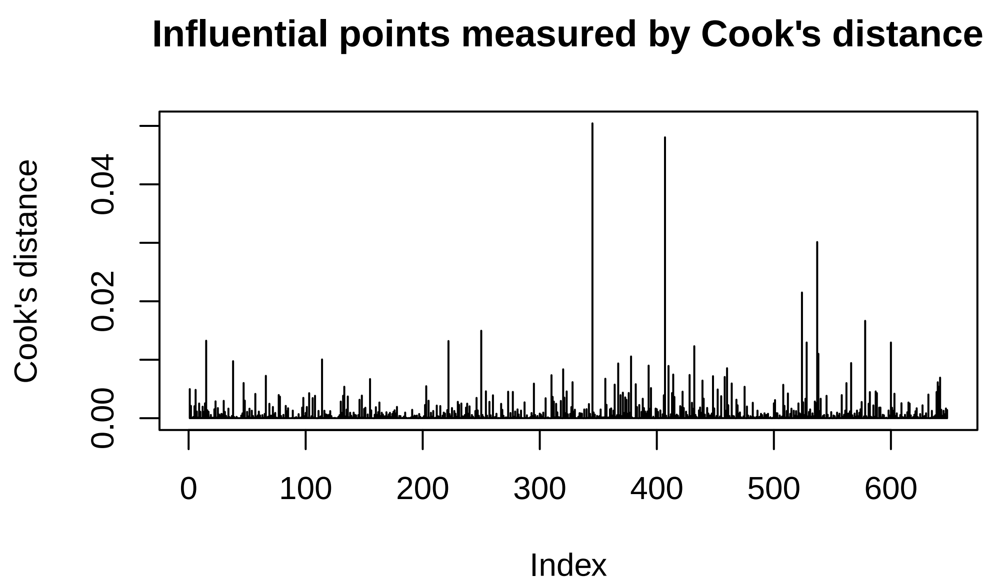

Q−Q Residuals
Standardized residuals
Theoretical Quantiles
m(log_hours ~ Depression * Primary.streaming.service + Age + While.work
407
345
537
2
0
−2
−4
−3
−2
−1
0
1
2
3






```
# we can see that violations of homoscedasticity and normality have improved (no more strong
```

```r
# Cook's distance for linear regression (RQ1)
cooks <- cooks.distance(model_rq1_log)

plot(cooks, type = "h",
     main = "Influential points measured by Cook’s distance",
     ylab = "Cook's distance",
     xlab = "Index")
```




```r
# Identify influential observations
which(cooks > 0.04)
```

```
345 407
345 407
```

```r
# Removing the three main outliers shown in the diagnostic plots
music_data_1_subset <- music_data_1[!(rownames(music_data_1) %in% c("407", "345", "537")), ]

# Refitting the model
# run log-transform
model_rq1_log_final <- lm(
  log_hours ~ Depression * Primary.streaming.service +
```

```r
    Age +
    While.working +
    Instrumentalist +
    Composer +
    Exploratory +
    Foreign.languages +
    Anxiety +
    Insomnia +
    OCD +
    Fav.genre +
    Music.effects,
  data = music_data_1_subset
)

summary(model_rq1_log_final)
```

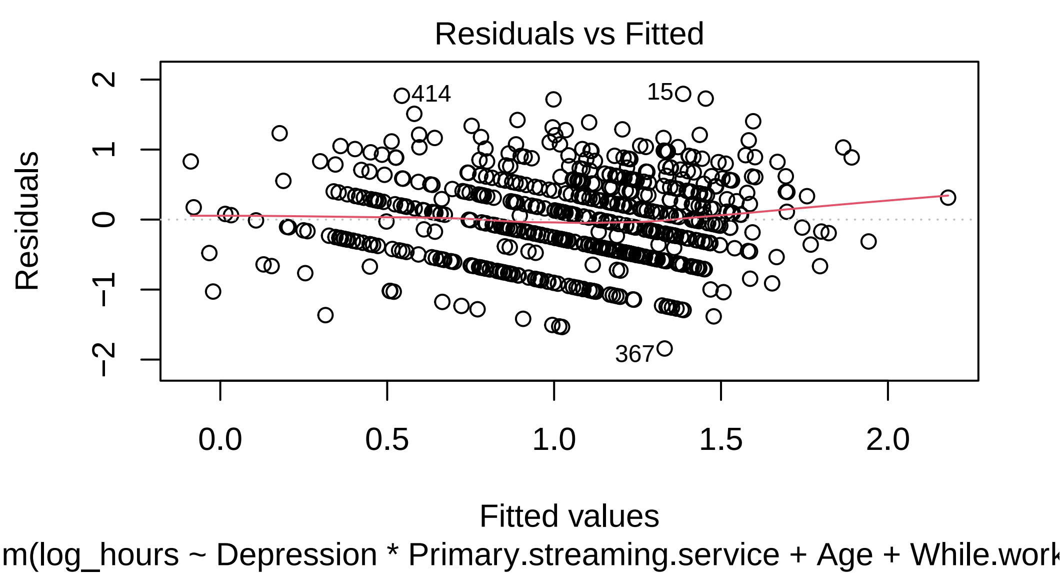
```

Call:
lm(formula = log_hours ~ Depression * Primary.streaming.service +
    Age + While.working + Instrumentalist + Composer + Exploratory +
    Foreign.languages + Anxiety + Insomnia + OCD + Fav.genre +
    Music.effects, data = music_data_1_subset)

Residuals:
    Min      1Q  Median      3Q     Max
-1.8419 -0.4221 -0.0150  0.4023  1.7956

Coefficients:
                                                              Estimate
(Intercept)                                                 -0.1327805
Depression                                                   0.1059303
Primary.streaming.serviceOther streaming service             0.6964129
Primary.streaming.servicePandora                             0.7757835
Primary.streaming.serviceSpotify                             0.5766650
Primary.streaming.serviceYouTube Music                       0.6249309
Age                                                         -0.0053219
While.workingYes                                             0.6047737
InstrumentalistYes                                          -0.1600362
ComposerYes                                                  0.1522387
ExploratoryYes                                               0.0715536
Foreign.languagesYes                                         0.0602961
Anxiety                                                      0.0004884
```

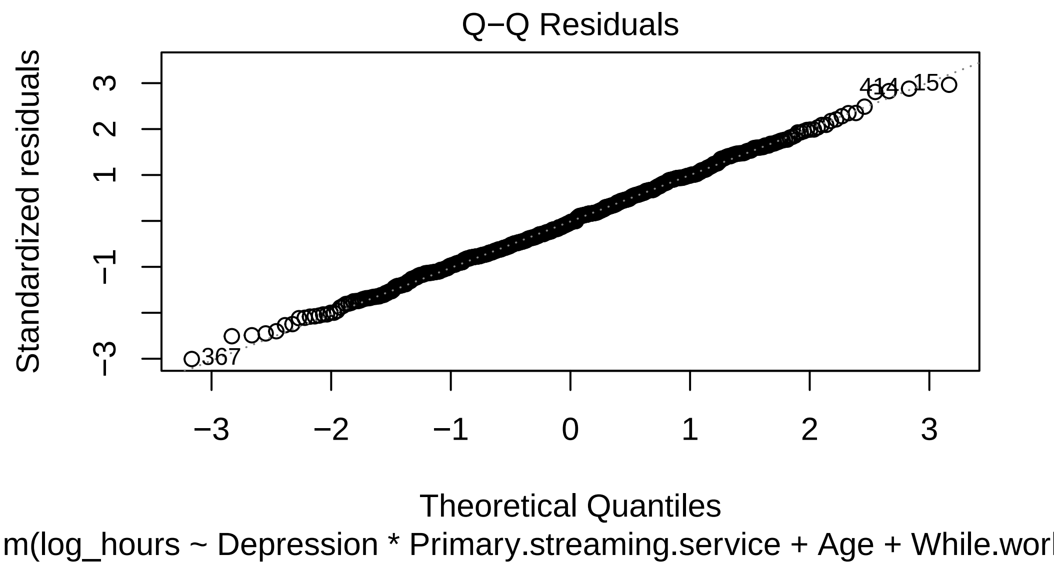
```
Insomnia                                                     0.0264703
OCD                                                          0.0186144
Fav.genreCountry                                             0.1492108
Fav.genreEDM                                                 0.2479763
Fav.genreFolk                                                0.0943413
Fav.genreGospel                                              0.0598351
Fav.genreHip hop                                             0.1344898
Fav.genreJazz                                                0.4032782
Fav.genreK pop                                              -0.0723137
Fav.genreLatin                                               0.9537137
Fav.genreLofi                                                0.0599377
Fav.genreMetal                                               0.1318610
Fav.genrePop                                                -0.1174466
Fav.genreR&B                                                 0.1280786
Fav.genreRap                                                 0.2834188
Fav.genreRock                                                0.0966310
Fav.genreVideo game music                                   -0.2663141
Music.effectsNo effect                                       0.0169920
Music.effectsWorsen                                         -0.1057281
Depression:Primary.streaming.serviceOther streaming service -0.1654575
Depression:Primary.streaming.servicePandora                 -0.2074562
Depression:Primary.streaming.serviceSpotify                 -0.1030834
Depression:Primary.streaming.serviceYouTube Music           -0.1401686
                                                            Std. Error t value
(Intercept)                                                  0.2366303  -0.561
Depression                                                   0.0321673   3.293
Primary.streaming.serviceOther streaming service             0.2447379   2.846
Primary.streaming.servicePandora                             0.3545808   2.188
Primary.streaming.serviceSpotify                             0.1971559   2.925
Primary.streaming.serviceYouTube Music                       0.2155796   2.899
Age                                                          0.0026183  -2.033
While.workingYes                                             0.0645942   9.363
InstrumentalistYes                                           0.0626386  -2.555
ComposerYes                                                  0.0754331   2.018
ExploratoryYes                                               0.0612935   1.167
Foreign.languagesYes                                         0.0535436   1.126
Anxiety                                                      0.0114229   0.043
Insomnia                                                     0.0091793   2.884
OCD                                                          0.0095310   1.953
Fav.genreCountry                                             0.1677984   0.889
Fav.genreEDM                                                 0.1494639   1.659
Fav.genreFolk                                                0.1613934   0.585
Fav.genreGospel                                              0.3166407   0.189
```

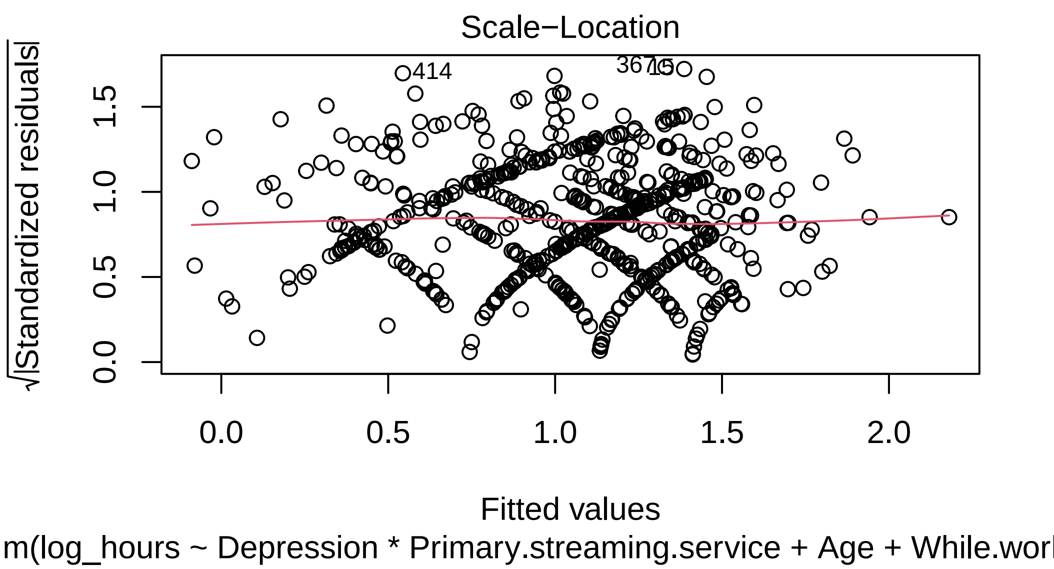
```
Fav.genreHip hop                                          0.1525111   0.882
Fav.genreJazz                                             0.1747707   2.307
Fav.genreK pop                                            0.1754936  -0.412
Fav.genreLatin                                            0.4569177   2.087
Fav.genreLofi                                             0.2273208   0.264
Fav.genreMetal                                            0.1255542   1.050
Fav.genrePop                                              0.1204029  -0.975
Fav.genreR&B                                              0.1534067   0.835
Fav.genreRap                                              0.1755902   1.614
Fav.genreRock                                             0.1147712   0.842
Fav.genreVideo game music                                 0.1561979  -1.705
Music.effectsNo effect                                    0.0628217   0.270
Music.effectsWorsen                                       0.1751111  -0.604
Depression:Primary.streaming.serviceOther streaming service  0.0430413  -3.844
Depression:Primary.streaming.servicePandora               0.0665342  -3.118
Depression:Primary.streaming.serviceSpotify               0.0333559  -3.090
Depression:Primary.streaming.serviceYouTube Music         0.0382832  -3.661
                                                          Pr(>|t|)
(Intercept)                                               0.574915
Depression                                                0.001048 **
Primary.streaming.serviceOther streaming service          0.004583 **
Primary.streaming.servicePandora                          0.029057 *
Primary.streaming.serviceSpotify                          0.003574 **
Primary.streaming.serviceYouTube Music                    0.003880 **
Age                                                       0.042531 *
While.workingYes                                           < 2e-16 ***
InstrumentalistYes                                        0.010863 *
ComposerYes                                               0.044009 *
ExploratoryYes                                            0.243509
Foreign.languagesYes                                      0.260561
Anxiety                                                   0.965911
Insomnia                                                  0.004069 **
OCD                                                       0.051273 .
Fav.genreCountry                                          0.374232
Fav.genreEDM                                              0.097609 .
Fav.genreFolk                                             0.559072
Fav.genreGospel                                           0.850181
Fav.genreHip hop                                          0.378213
Fav.genreJazz                                             0.021363 *
Fav.genreK pop                                            0.680441
Fav.genreLatin                                            0.037278 *
Fav.genreLofi                                             0.792123
Fav.genreMetal                                            0.294028
```

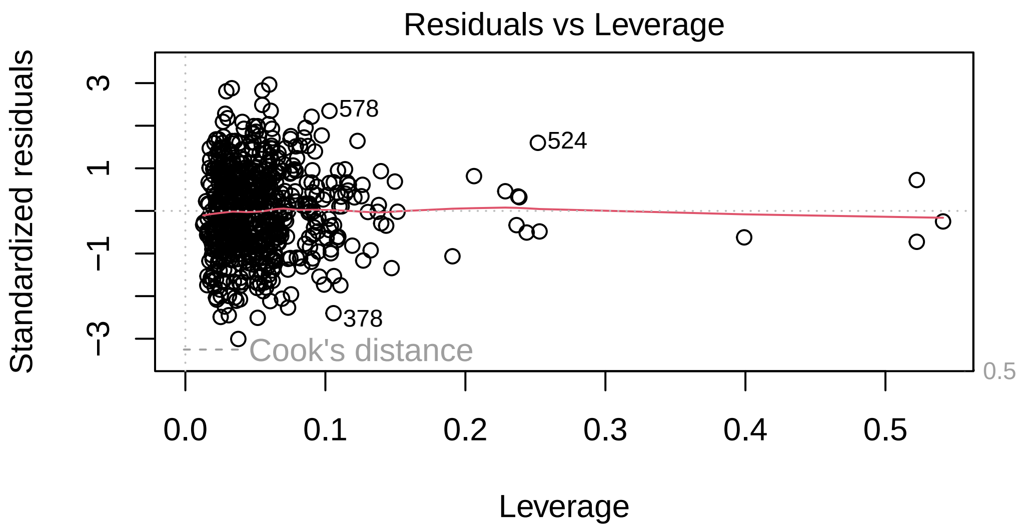
```
Fav.genrePop                                                     0.329726
Fav.genreR&B                                                     0.404104
Fav.genreRap                                                     0.107025
Fav.genreRock                                                    0.400149
Fav.genreVideo game music                                        0.088708 .
Music.effectsNo effect                                           0.786883
Music.effectsWorsen                                              0.546217
Depression:Primary.streaming.serviceOther streaming service 0.000134 ***
Depression:Primary.streaming.servicePandora                      0.001906 **
Depression:Primary.streaming.serviceSpotify                      0.002090 **
Depression:Primary.streaming.serviceYouTube Music                0.000273 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.6246 on 609 degrees of freedom
Multiple R-squared:  0.2623,    Adjusted R-squared:  0.2199
F-statistic: 6.185 on 35 and 609 DF,  p-value: < 2.2e-16
```

```r
# Diagnostic plots for the final model, after log transformation and removal of outliers
plot(model_rq1_log_final)
```




Q−Q Residuals
Standardized residuals
Theoretical Quantiles
m(log_hours ~ Depression * Primary.streaming.service + Age + While.work
−3
−1
1
2
3
−3
−2
−1
0
1
2
3
367
414
15


Scale−Location
√|Standardized residuals|
Fitted values
m(log_hours ~ Depression * Primary.streaming.service + Age + While.work
0.0
0.5
1.0
1.5
0.0
0.5
1.0
1.5
2.0
414
367
15



m(log_hours ~ Depression * Primary.streaming.service + Age + While.work

Table with Key Regression Coefficients

```r
library(broom)
library(kableExtra)
```

```

Attaching package: 'kableExtra'


The following object is masked from 'package:dplyr':

    group_rows
```

```r
library(dplyr)

tbl_rq1_sig <- tidy(
  model_rq1_log_final,
  conf.int = TRUE
) %>%
  filter(p.value < 0.05) %>%           # significant predictors only
```

```r
  mutate(
    estimate  = round(estimate, 3),
    std.error = round(std.error, 3),
    conf.low  = round(conf.low, 3),
    conf.high = round(conf.high, 3),
    p.value   = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value))
  ) %>%
  select(term, estimate, std.error, conf.low, conf.high, p.value)

kable(tbl_rq1_sig,
      caption = "Significant Predictors from RQ1 Linear Model",
      booktabs = TRUE) %>%
  kable_styling(full_width = FALSE)
```

Table 1: Significant Predictors from RQ1 Linear Model

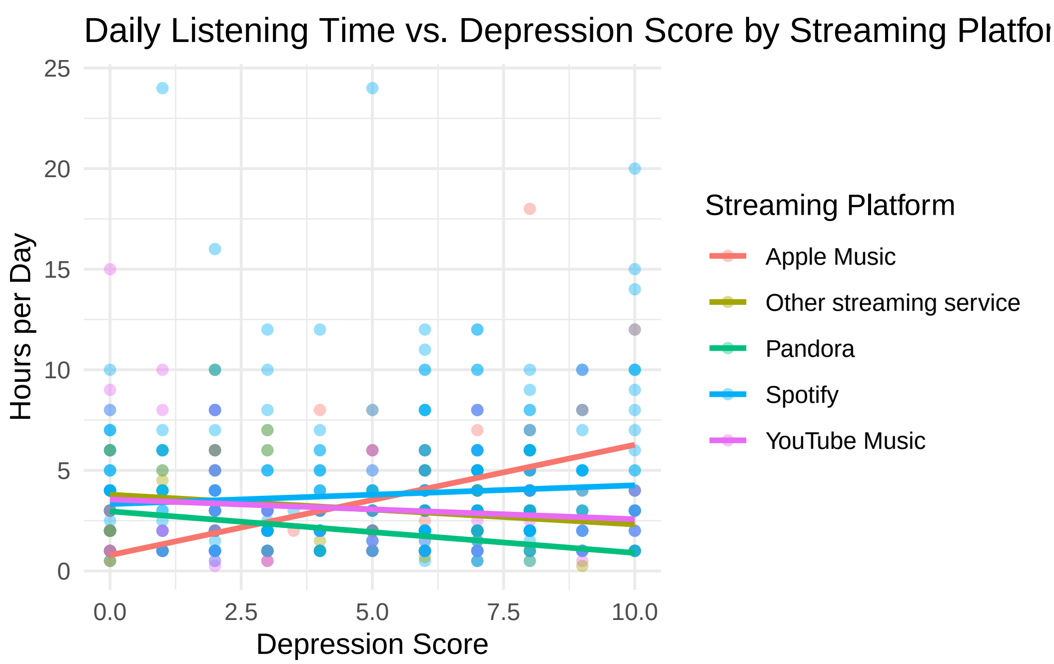
term	estimate	std.error	conf.low	conf.high	p.
Depression	0.106	0.032	0.043	0.169	0.
Primary.streaming.serviceOther streaming service	0.696	0.245	0.216	1.177	0.
Primary.streaming.servicePandora	0.776	0.355	0.079	1.472	0.
Primary.streaming.serviceSpotify	0.577	0.197	0.189	0.964	0.
Primary.streaming.serviceYouTube Music	0.625	0.216	0.202	1.048	0.
Age	-0.005	0.003	-0.010	0.000	0.
While.workingYes	0.605	0.065	0.478	0.732	<
InstrumentalistYes	-0.160	0.063	-0.283	-0.037	0.
ComposerYes	0.152	0.075	0.004	0.300	0.
Insomnia	0.026	0.009	0.008	0.044	0.
Fav.genreJazz	0.403	0.175	0.060	0.747	0.
Fav.genreLatin	0.954	0.457	0.056	1.851	0.
Depression:Primary.streaming.serviceOther streaming service	-0.165	0.043	-0.250	-0.081	<
Depression:Primary.streaming.servicePandora	-0.207	0.067	-0.338	-0.077	0.
Depression:Primary.streaming.serviceSpotify	-0.103	0.033	-0.169	-0.038	0.
Depression:Primary.streaming.serviceYouTube Music	-0.140	0.038	-0.215	-0.065	<

Predicted Listening Time vs Depression by Streaming Platform

```
ggplot(music_data_1_subset,
       aes(x = Depression,
           y = Hours.per.day,
           color = Primary.streaming.service)) +
  geom_point(alpha = 0.4) +
  geom_smooth(method = "lm", se = FALSE, size = 1) +
  labs(
    title = "Daily Listening Time vs. Depression Score by Streaming Platform",
    x = "Depression Score",
    y = "Hours per Day",
    color = "Streaming Platform"
  ) +
  theme_minimal()
```

```
Warning: Using `size` aesthetic for lines was deprecated in ggplot2 3.4.0.
i Please use `linewidth` instead.
```

```
`geom_smooth()` using formula = 'y ~ x'
```



Research Question 2

```r
# Question 2: Does the number of hours spent listening to music each day increase the likelil

# this will be the dataset used for our first research question
music_data_2 <- music_data_clean

# create high_depression as 1 if Depression >= threshold, 0 otherwise
music_data_2$high_depression <- ifelse(music_data_2$Depression >= 6, 1, 0)

# Fit logistic regression including all relevant predictors
model2 <- glm(
  high_depression ~ Hours.per.day +
    Age +
    Instrumentalist +
    Composer +
    Anxiety +
    Insomnia +
    OCD +
    Fav.genre +
    Music.effects,
  data = music_data_2,
  family = binomial
)

# View model summary
summary(model2)
```

```
Call:
glm(formula = high_depression ~ Hours.per.day + Age + Instrumentalist +
    Composer + Anxiety + Insomnia + OCD + Fav.genre + Music.effects,
    family = binomial, data = music_data_2)

Coefficients:
                           Estimate Std. Error z value Pr(>|z|)
(Intercept)               -2.491848   0.517394  -4.816 1.46e-06 ***
Hours.per.day              0.047603   0.032242   1.476   0.1398
Age                       -0.012897   0.008612  -1.498   0.1342
InstrumentalistYes        -0.062479   0.215923  -0.289   0.7723
ComposerYes                0.118458   0.259171   0.457   0.6476
```

```
Anxiety                      0.335503   0.038639   8.683  < 2e-16 ***
Insomnia                     0.155857   0.030598   5.094 3.51e-07 ***
OCD                         -0.026242   0.032840  -0.799   0.4243
Fav.genreCountry            -0.562543   0.590770  -0.952   0.3410
Fav.genreEDM                 0.161755   0.532781   0.304   0.7614
Fav.genreFolk               -0.233364   0.532752  -0.438   0.6614
Fav.genreGospel             -1.622996   1.304024  -1.245   0.2133
Fav.genreHip hop             0.933621   0.538995   1.732   0.0832 .
Fav.genreJazz               -0.328074   0.614412  -0.534   0.5934
Fav.genreK pop              -0.900573   0.609671  -1.477   0.1396
Fav.genreLatin               0.010090   1.584461   0.006   0.9949
Fav.genreLofi                0.681680   0.825330   0.826   0.4088
Fav.genreMetal               0.160046   0.421504   0.380   0.7042
Fav.genrePop                -0.419607   0.406713  -1.032   0.3022
Fav.genreR&B                -0.314000   0.549763  -0.571   0.5679
Fav.genreRap                -0.468758   0.623842  -0.751   0.4524
Fav.genreRock                0.265476   0.381712   0.695   0.4867
Fav.genreVideo game music -0.563356   0.494857  -1.138   0.2549
Music.effectsNo effect       0.104103   0.214565   0.485   0.6275
Music.effectsWorsen          0.955706   0.598303   1.597   0.1102
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

(Dispersion parameter for binomial family taken to be 1)

    Null deviance: 992.90  on 717  degrees of freedom
Residual deviance: 789.16  on 693  degrees of freedom
AIC: 839.16

Number of Fisher Scoring iterations: 4
```

```r
# Coefficients on the odds scale
exp(coef(model2))
```

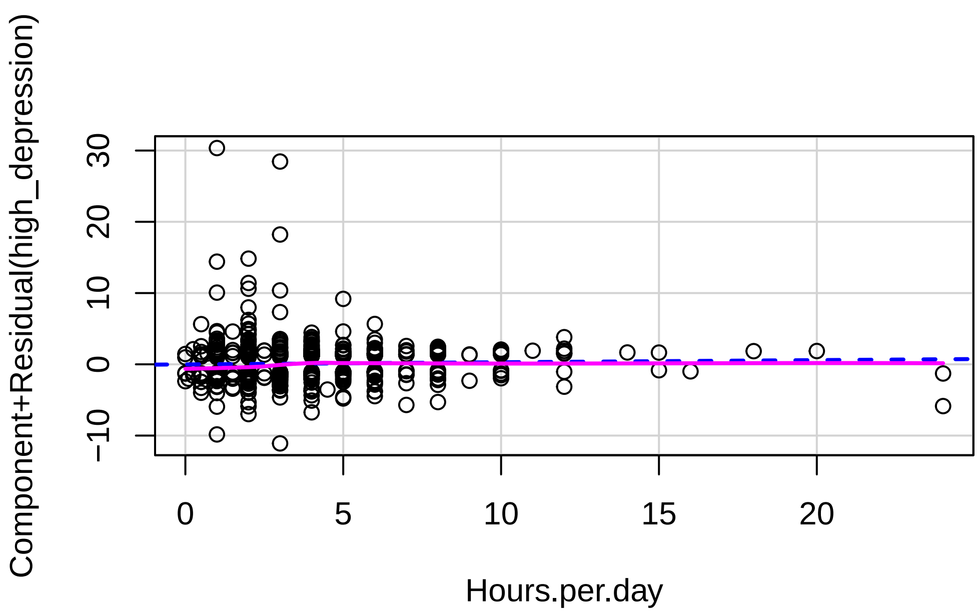
```
          (Intercept)          Hours.per.day                    Age
           0.08275692             1.04875372             0.98718560
    InstrumentalistYes            ComposerYes                Anxiety
           0.93943309             1.12575950             1.39864318
             Insomnia                    OCD       Fav.genreCountry
           1.16865869             0.97409978             0.56975853
         Fav.genreEDM          Fav.genreFolk        Fav.genreGospel
           1.17557231             0.79186532             0.19730668
```

```
        Fav.genreHip hop             Fav.genreJazz             Fav.genreK pop
              2.54370243                0.72031004                0.40633668
          Fav.genreLatin             Fav.genreLofi            Fav.genreMetal
              1.01014147                1.97719690                1.17356536
            Fav.genrePop              Fav.genreR&B              Fav.genreRap
              0.65730505                0.73051889                0.62577909
           Fav.genreRock Fav.genreVideo game music    Music.effectsNo effect
              1.30405221                0.56929513                1.10971472
     Music.effectsWorsen
              2.60050537
```

Diagnostics RQ2

Partial residual plot

```r
# Partial residual plot for our main predictor (Hours.per.day)

# This checks linearity of Hours.per.day with log-odds.
crPlot(model2, "Hours.per.day")
```



Checking for multicollinearity

```r
vif(model2)
```

```
                    GVIF Df GVIF^(1/(2*Df))
Hours.per.day   1.104248  1        1.050832
Age             1.182510  1        1.087433
Instrumentalist 1.351533  1        1.162555
Composer        1.245202  1        1.115886
Anxiety         1.172575  1        1.082855
Insomnia        1.106988  1        1.052135
OCD             1.172438  1        1.082792
Fav.genre       1.577869 15        1.015319
Music.effects   1.099217  2        1.023931
```

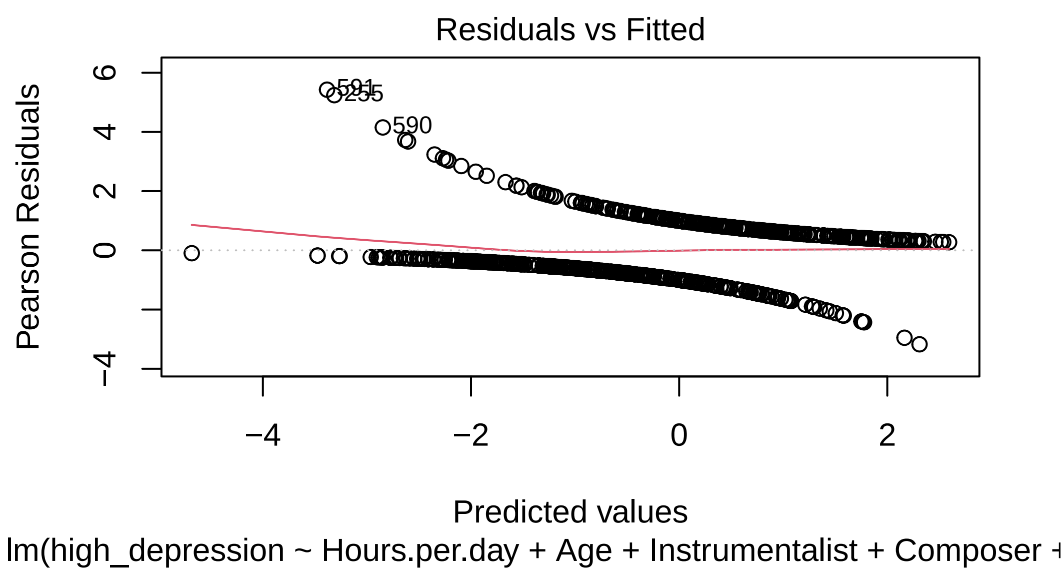
Goodness-of-fit (Hosmer-Lemeshow test)

```r
# This test compares observed and expected frequencies of the outcome.
ResourceSelection::hoslem.test(model2$y, fitted(model2))
```

```

    Hosmer and Lemeshow goodness of fit (GOF) test

data:  model2$y, fitted(model2)
X-squared = 11.722, df = 8, p-value = 0.1641
```

Pearson residuals vs fitted plot

```r
plot(model2, which = 1)
```



Checking for high leverage points

```
hatvalues <- hatvalues(model2)
which(hatvalues > (2*mean(hatvalues)))
```

```
  2   9  12  13  16  50  57 121 137 147 207 226 233 237 239 243 270 287 292 313
  2   9  12  13  16  50  57 121 137 147 207 226 233 237 239 243 270 287 292 313
339 344 346 348 369 386 424 427 430 438 443 464 469 470 479 484 515 520 553 564
339 344 346 348 369 386 424 427 430 438 443 464 469 470 479 484 515 520 553 564
579 580 583 593 623 641 657 666 669 697
579 580 583 593 623 641 657 666 669 697
```

Confusion Matrix

```
probs <- predict(model2, type = "response")

pred_class <- ifelse(probs > 0.5, 1, 0)
```

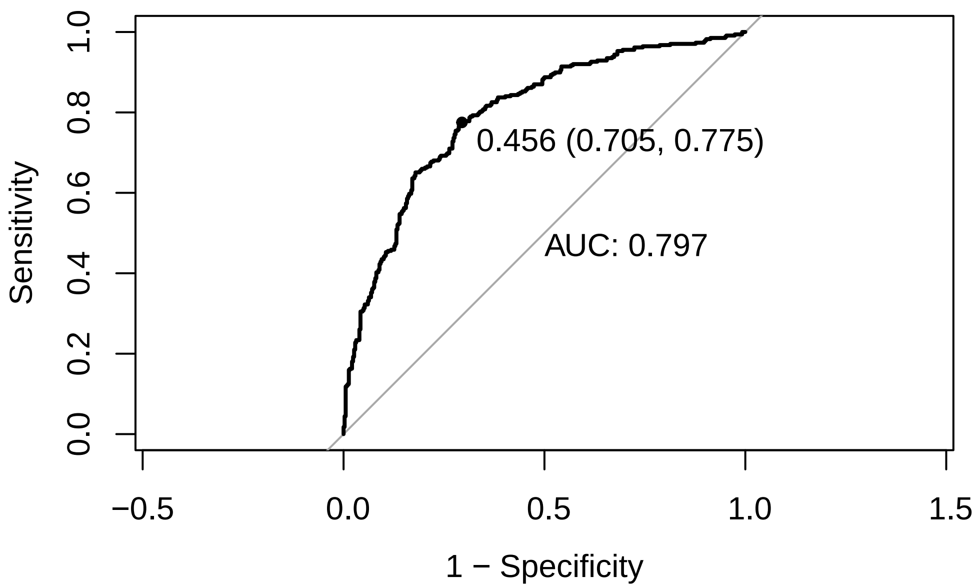
```r
pred_class <- factor(pred_class, levels = c(0,1))

caret::confusionMatrix(
  table(pred_class, music_data_2$high_depression),
  positive = "1",
  mode = "everything"
)
```

```
Confusion Matrix and Statistics


pred_class   0   1
         0 277  95
         1 103 243

               Accuracy : 0.7242
                 95% CI : (0.69, 0.7566)
    No Information Rate : 0.5292
    P-Value [Acc > NIR] : <2e-16

                  Kappa : 0.4473

 Mcnemar's Test P-Value : 0.6189

            Sensitivity : 0.7189
            Specificity : 0.7289
         Pos Pred Value : 0.7023
         Neg Pred Value : 0.7446
              Precision : 0.7023
                 Recall : 0.7189
                     F1 : 0.7105
             Prevalence : 0.4708
         Detection Rate : 0.3384
   Detection Prevalence : 0.4819
      Balanced Accuracy : 0.7239

       'Positive' Class : 1
```

ROC Curve and AUC

```
pROC::roc(music_data_2$high_depression, probs,
          print.thres = "best",
          print.auc = TRUE,
          legacy.axes = TRUE,
          plot = TRUE)
```

```
Setting levels: control = 0, case = 1

Setting direction: controls < cases
```




```
Call:
roc.default(response = music_data_2$high_depression, predictor = probs,     plot = TRUE, pri

Data: probs in 380 controls (music_data_2$high_depression 0) < 338 cases (music_data_2$high_d
Area under the curve: 0.7968
```

Updated Confusion Matrix with "best" Threshold

```r
probs <- predict(model2, type = "response")

pred_class <- ifelse(probs > 0.456, 1, 0)
pred_class <- factor(pred_class, levels = c(0,1))

caret::confusionMatrix(
  table(pred_class, music_data_2$high_depression),
  positive = "1",
  mode = "everything"
)
```

```
Confusion Matrix and Statistics


pred_class   0   1
         0 268  76
         1 112 262

               Accuracy : 0.7382
                 95% CI : (0.7044, 0.77)
    No Information Rate : 0.5292
    P-Value [Acc > NIR] : < 2e-16

                  Kappa : 0.4776

 Mcnemar's Test P-Value : 0.01069

            Sensitivity : 0.7751
            Specificity : 0.7053
         Pos Pred Value : 0.7005
         Neg Pred Value : 0.7791
              Precision : 0.7005
                 Recall : 0.7751
                     F1 : 0.7360
             Prevalence : 0.4708
         Detection Rate : 0.3649
   Detection Prevalence : 0.5209
      Balanced Accuracy : 0.7402
```

```
'Positive' Class : 1
```

Reduced Model Comparison (no Hours.per.day)

```r
# reduced model without Hours.per.day
model2_red <- glm(
  high_depression ~
    Age +
    Instrumentalist +
    Composer +
    Anxiety +
    Insomnia +
    OCD +
    Fav.genre +
    Music.effects,
  data = music_data_2,
  family = binomial
)


# Deviance test
anova(model2_red, model2, test="Chisq")
```

```
Analysis of Deviance Table

Model 1: high_depression ~ Age + Instrumentalist + Composer + Anxiety +
    Insomnia + OCD + Fav.genre + Music.effects
Model 2: high_depression ~ Hours.per.day + Age + Instrumentalist + Composer +
    Anxiety + Insomnia + OCD + Fav.genre + Music.effects
  Resid. Df Resid. Dev Df Deviance Pr(>Chi)
1       694     791.39
2       693     789.16  1    2.229   0.1354
```

```r
AIC(model2_red)
```

```
[1] 839.3908
```

```r
AIC(model2)
```

```
[1] 839.1618
```

** Confidence Intervals and Odds Ratios**

```r
exp(cbind(OR = coef(model2), confint(model2)))
```

```
Waiting for profiling to be done...
```

```
                                OR       2.5 %     97.5 %
(Intercept)             0.08275692 0.029458620  0.2246119
Hours.per.day           1.04875372 0.985376521  1.1186433
Age                     0.98718560 0.970355398  1.0037732
InstrumentalistYes      0.93943309 0.614457810  1.4341172
ComposerYes             1.12575950 0.677096843  1.8736563
Anxiety                 1.39864318 1.298747929  1.5114573
Insomnia                1.16865869 1.101200483  1.2417096
OCD                     0.97409978 0.913205223  1.0388571
Fav.genreCountry        0.56975853 0.174224432  1.7924863
Fav.genreEDM            1.17557231 0.414376005  3.3662330
Fav.genreFolk           0.79186532 0.277383692  2.2584421
Fav.genreGospel         0.19730668 0.008112022  1.9886755
Fav.genreHip hop        2.54370243 0.897859848  7.4883957
Fav.genreJazz           0.72031004 0.214000862  2.4151809
Fav.genreK pop          0.40633668 0.118808517  1.3206120
Fav.genreLatin          1.01014147 0.032147769 32.4242542
Fav.genreLofi           1.97719690 0.419048116 11.4650789
Fav.genreMetal          1.17356536 0.514383845  2.6961921
Fav.genrePop            0.65730505 0.295924805  1.4639383
Fav.genreR&B            0.73051889 0.244760894  2.1279338
Fav.genreRap            0.62577909 0.177937616  2.0891612
Fav.genreRock           1.30405221 0.618178484  2.7736425
Fav.genreVideo game music 0.56929513 0.213595710  1.4954360
Music.effectsNo effect  1.10971472 0.728565498  1.6915840
Music.effectsWorsen     2.60050537 0.841506761  9.1406607
```

Final number of rows for each subset of data (RQ1 and RQ2)

```r
nrow(music_data_1_subset)
```

```
[1] 645
```

```r
nrow(music_data_2)
```

```
[1] 718
```

Condensed Odds Ratios Table for RQ2 (Only Key Predictors)

```r
tbl_rq2_condensed <- tidy(
  model2,
  exponentiate = TRUE,
  conf.int = TRUE
) %>%
  filter(term %in% c("Hours.per.day", "Anxiety", "Insomnia")) %>%  # key predictors
  mutate(
    estimate  = round(estimate, 2),
    conf.low  = round(conf.low, 2),
    conf.high = round(conf.high, 2),
    p.value   = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value))
  ) %>%
  select(term, estimate, conf.low, conf.high, p.value)

kable(
  tbl_rq2_condensed,
  caption = "Key Predictors from Logistic Regression (Odds Ratios)",
  booktabs = TRUE
) %>%
  kable_styling(full_width = FALSE)
```

Table 2: Key Predictors from Logistic Regression (Odds Ratios)

term	estimate	conf.low	conf.high	p.value

Hours.per.day	1.05	0.99	1.12	0.140
Anxiety	1.40	1.30	1.51	<0.001
Insomnia	1.17	1.10	1.24	<0.001

```r
library(broom)
library(kableExtra)
library(dplyr)

tbl_rq1_sig <- tidy(
  model_rq1_log_final,
  conf.int = TRUE
) %>%
  filter(p.value < 0.05) %>%
  mutate(
    # exponentiate coefficients + CI
    exp_estimate = exp(estimate),
    exp_low      = exp(conf.low),
    exp_high     = exp(conf.high),

    # round all columns appropriately
    estimate     = round(estimate, 3),
    std.error    = round(std.error, 3),
    conf.low     = round(conf.low, 3),
    conf.high    = round(conf.high, 3),
    exp_estimate = round(exp_estimate, 3),
    exp_low      = round(exp_low, 3),
    exp_high     = round(exp_high, 3),
    p.value      = ifelse(p.value < 0.001, "<0.001", sprintf("%.3f", p.value))
  ) %>%
  select(
    term,
    estimate,
    exp_estimate,
    std.error,
    conf.low,
    conf.high,
    exp_low,
    exp_high,
    p.value
  )

kable(
```

```
  tbl_rq1_sig,
  booktabs = TRUE,
  col.names = c(
    "Term",
    "Estimate (log scale)",
    "Exp(Estimate)",
    "Std Error",
    "2.5% CI",
    "97.5% CI",
    "Exp(2.5% CI)",
    "Exp(97.5% CI)",
    "p-value"
  )
) %>%
  kable_styling(full_width = FALSE, font_size = 9)
```

Term	Estimate (log scale)	Exp(Estimate)	Std Error	2.5% CI
Depression	0.106	1.112	0.032	0.043
Primary.streaming.serviceOther streaming service	0.696	2.007	0.245	0.216
Primary.streaming.servicePandora	0.776	2.172	0.355	0.079
Primary.streaming.serviceSpotify	0.577	1.780	0.197	0.189
Primary.streaming.serviceYouTube Music	0.625	1.868	0.216	0.202
Age	-0.005	0.995	0.003	-0.010
While.workingYes	0.605	1.831	0.065	0.478
InstrumentalistYes	-0.160	0.852	0.063	-0.283
ComposerYes	0.152	1.164	0.075	0.004
Insomnia	0.026	1.027	0.009	0.008
Fav.genreJazz	0.403	1.497	0.175	0.060
Fav.genreLatin	0.954	2.595	0.457	0.056
Depression:Primary.streaming.serviceOther streaming service	-0.165	0.848	0.043	-0.250
Depression:Primary.streaming.servicePandora	-0.207	0.813	0.067	-0.338
Depression:Primary.streaming.serviceSpotify	-0.103	0.902	0.033	-0.169
Depression:Primary.streaming.serviceYouTube Music	-0.140	0.869	0.038	-0.215

Table 1: Significant Predictors from RQ1 Linear Model (Original and Exponentiated Coefficients).